

Global Memory

CXMT Corp

Rating

Outperform

Price Target

688825.CH

70.00 CNY



Mark Li
+852 2123 2645
mark.li@bernsteinsg.com



Edward Hou, CFA
+852 2123 2623
edward.hou@bernsteinsg.com



Yipin Cai, CFA
+852 2123 2669
yipin.cai@bernsteinsg.com

China Next Winners: CXMT - Redrawing the Memory Map - Initiating with Outperform

This report is part of a series on China Next Winners in which we identify the next generation of industries and corporate winners in China.

Initiate coverage on CXMT (ChangXin Memory Technologies) with Outperform, as 1) cycle remains elevated, 2) localization & HBM make CXMT resilient even when the current cycle turns, & 3) valuation below Chinese peers'.

China's biggest DRAM maker and is gaining share rapidly. We model CXMT to defy export control to grow capacity/revenue/EPS by 75%/55%/50% from 2026 to 2028.

Localization & HBM will dampen cyclicity. CXMT's tech is c. 3-4 years behind but good enough for conventional server DRAM. No HBM revenue yet, but meaningful in 2028 with profitability protected by policy. We model the sector to "normalize" to lower but healthy margins in 2028, but localization demand & HBM will make CXMT's dip even smaller.

DUV can keep CXMT going for a while. EUV is a bottleneck & innovations (e.g. 4F² & 3D DRAM) are being researched to circumvent it. Hard to know if they will succeed but DUV should be able to support CXMT for likely another 3-5 years.

Above consensus in 2026/27 & CXMT is notably cheaper than A-share peers. Our 2028 EPS is in line but our 2026-27 revenue/EPS is 20-30%/40-50% above consensus. Shares unlocking in Jan & July 2027 is a risk, but attractive valuation & above-consensus forecast lead us to rate CXMT **Outperform**.

Investment Implications

We rate CXMT **Outperform**, with price target of **CNY 70** on 10x of forward Q5-Q8 EPS.

Reported EPS	F25A	F26E	F27E
688825.CH (CNY)	0.03	4.04	6.82

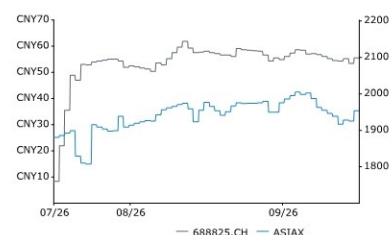
Financials	F25A	F26E	F27E	CAGR
Revenues (M)	61,799	452,851	749,441	248.2%
BV/Share (M)	0.94	5.65	12.47	263.6%
CapEx (M)	49,739	81,291	141,540	68.7%
Gross Margin (%)	41.0	88.1	90.5	--
EBIT (M)	10,655	377,977	642,130	676.3%
Net Income Margin (%)	3.0	57.3	61.8	--
ROE (%)	3.8	118.0	75.3	--

Close Date	18 Sep 2026
688825.CH Close Price (CNY)	55.54
Price Target (CNY)	70.00
Upside/(Downside)	26%
52-Week Range	61.80/38.11
ASIAx	1,953.49
FYE	Dec
Div Yield	NA
Market Cap (CNY) (B)	3,770.28
EV (CNY) (B)	3,867.56

Performance	YTD	1M	6M	12M
Absolute (%)	NA	(6.2)	NA	NA
ASIAx (%)	19.4	(1.2)	15.4	23.8
Relative (%)	NA	(5.0)	NA	NA

Source: Bloomberg, Bernstein estimates and analysis.

Price Performance, 1YR



Valuation Metrics	F25A	F26E	F27E
Reported P/E (x)	N/M	13.8	8.1
P/B (x)	58.9	9.8	4.5
Div Yield (%)	0.0	0.0	0.0
EV/EBITDA (x)	102.4	9.4	5.6

Source: Bloomberg, Bernstein estimates and analysis.

See the Disclosure Appendix of this report for required disclosures, analyst certifications and other important information. Alternatively, visit our [Global Research Disclosure Website](#).

First Published: 21 Sep 2026 08:30 UTC Completion Date: 18 Sep 2026 16:01 UTC

www.bernsteinresearch.com

Table Of Contents

Executive Summary.....	2
Aggressive capex & capacity expansion.....	3
Good enough technology for conventional DRAM.....	5
Can export control keep CXMT from growing & from AI (i.e. HBM)?.....	9
Competitive only in China & in low-end segments?.....	16
Can China be self-sufficient in DRAM?.....	18
Financial projection combining both cyclical & idiosyncratic perspectives.....	22
Valuation, rating & risk.....	27

DETAILS

EXECUTIVE SUMMARY

We're positive on CXMT because 1) the industry is still amid an elevated cycle, 2) localization demand & catching-up in HBM will make CXMT fare better than peers even when cycle turns, & 3) CXMT is cheaper than Chinese peers. We have been discussing the memory industry cycle, & this report will focus on CXMT in expansion, technology, export control, addressable market, financial performance and valuation.

High capex to surpass Micron in wafer capacity in 2028. Considering CXMT's cash position & China's self-sufficiency need, we expect CXMT to spend more than consensus, possibly with new fabs announced later, & project its capex to rise to CNY 81/142/236B (US\$12/21/35B) in 2026/27/28 ([Exhibit 3](#)). Its wafer capacity is modeled to grow from 300K wafers/month (wpm) now to 530K in 4Q28 ([Exhibit 1](#)) & propel CXMT to surpass Micron in wafer capacity then ([Exhibit 2](#)). This however assumes the existing export control ([more here](#)) won't slow CXMT down (as no sign at all thus far) & won't become tighter either (as the US needs rare earth from China).

Lagging technologies but good enough for conventional DRAM. Teardowns found CXMT's bit density is 3-4 years behind ([Exhibit 6](#)), as its current technology "G4" is similar to 1z-1anm from others. Despite that, CXMT has been quickly lowering its cost & narrowing its gross margin gap vs. rivals' before the current cycle started ([Exhibit 15](#)). Further thanks to the cycle, CXMT's gross margin soared to 88% in 2Q26, as good as 87-89% from Samsung, SK hynix & Micron ([Exhibit 12](#)). CXMT is also good enough for demanding applications such as server, & had 27% of its revenue from server in 2025 ([Exhibit 27](#)) & very likely more now. The advancement from DDR4 to DDR5 is a success too ([Exhibit 9](#)), with DDR5 easily representing over 50% of its revenue now. [Xiaomi's recent announcement to adopt CXMT's LPDDR6 also marks its entry to the latest generation too.](#)

HBM is technologically challenging but demand & profitability is protected by policy. HBM demands speedier DRAM dies that CXMT's lagging technologies may find hard to deliver. TSV & packaging are also new to CXMT. More TSVs are needed in the future generations of HBM & export control limiting equipment capable of >3000 TSVs/die will thus become a mounting barrier additionally. Finally base die is shifting to advanced logic nodes that China is struggling too. All these imply low yield & high cost for CXMT, but policy requires Chinese customers have to buy HBM from CXMT, and should guarantee demand & sufficient profitability ([Exhibit 23](#)) for CXMT.

By expanding global bit/revenue share from 9%/10% this year to 12%/13% in 2028, CXMT can increase China's DRAM self-sufficiency from 42% to 58%. Considering the lower bit density, we discount CXMT's wafer share to estimate its bit share, but still project the bit share to rise to 12% in 2028 ([Exhibit 28](#)). As supply chain inputs & analysis suggest CXMT's price is similar to others', we model its revenue share to rise to 13% at the same time ([Exhibit 30](#)). Assuming China's goal is to fulfill the need of Chinese OEMs (e.g. including the need for Lenovo to sell PCs in Europe), we estimate CXMT will increase China's DRAM self-sufficiency from 42% now to 58% in 2028 ([Exhibit 33](#)).

CXMT's TAM includes a meaningful part of overseas market. Many assume geopolitical concerns will keep CXMT's memory only in China. We however highlight first that Chinese OEMs will install CXMT's memory in their end products & those end

products will be sold to overseas customers. Moreover, our checks also consistently find foreign PC, smartphone & consumer OEMs in general can accept the use of Chinese memory in products sold globally, likely except in the US & in sensitive segments for government or data center AI, etc.

No EUV but DUV may keep CXMT competitive for another 3-5 years. CXMT is trying to circumvent the need of EUV with 4F² & 3D DRAM ([Exhibit 24](#), [Exhibit 25](#)), but they won't happen meaningfully in our forecast horizon up to 2028. More importantly Micron produces DRAM without EUV up to 18nm ([Exhibit 7](#)), & CXMT likely can stretch DUV further. With that we believe DUV is sufficient to sustain CXMT for another 3-5 years.

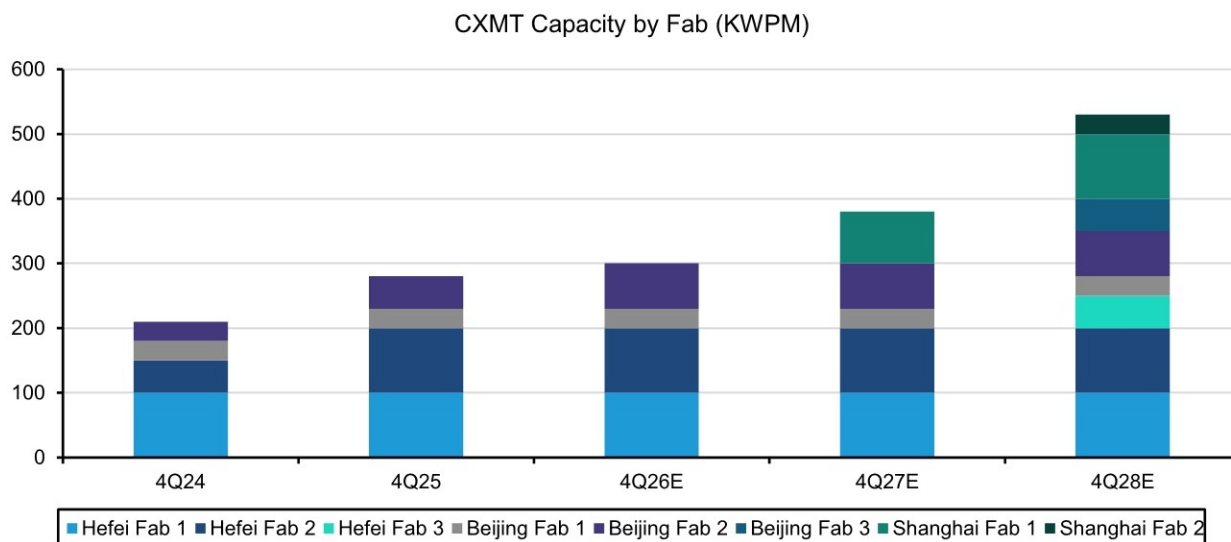
We model CXMT's EPS at CNY4/6.8/6 in 2026/27/28 ([Exhibit 44](#)). We expect the industry DRAM price to stay elevated in 2027 but begin to "normalize" in 2028. CXMT can't escape the cyclical, but should be more resilient as localization ensures better demand & HBM reaches notable volume in 2028. Our relative cyclical view makes us slightly below consensus in 2028 revenue & in line in EPS, but we are 20-30% above in revenue & 40-50% above in EPS in 2026 & 2027 ([Exhibit 46](#)). Notably minority interest is difficult to model & is a risk to our EPS forecast.

With valuation of 10x of Q5-Q8 EPS, we rate CXMT Outperform with price target of CNY 70. CXMT has pricier valuation than overseas peers, but that premium should stay as A-share tech stocks more than usually enjoy a premium over their overseas counterparts ([Exhibit 47](#), [Exhibit 48](#)). More importantly CXMT is surprisingly cheap vs. A-share semiconductor peers ([Exhibit 50](#)). For those worry about cycle and thus earnings two years from now, CXMT is also cheaper on 2-year forward basis ([Exhibit 51](#)). We caution investors that many shares will be freed from lock-up in Jan & July next year ([Exhibit 52](#)), but with attractive valuation & our above-consensus forecast, we rate CXMT Outperform with price target of CNY 70 on 10x of Q5-Q8 EPS. That implies c. 26% upside as of Sep 18th.

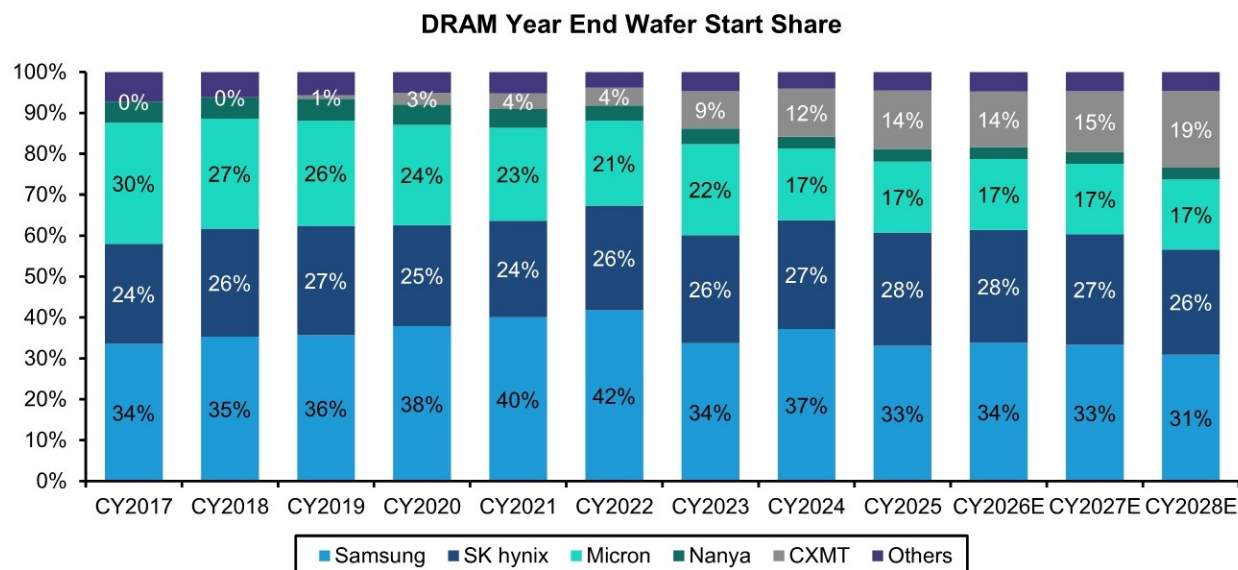
AGGRESSIVE CAPEX & CAPACITY EXPANSION

With c. 75% expansion from now to 4Q28, CXMT is expected to have c. 20% of the global capacity & surpass Micron in wafer capacity in 2028.

- CXMT currently has capacity of around 300K wafers per month (wpm). Existing fabs in Hefei and Beijing likely have largely reached full capacity, but new fabs are being built and should start to contribute new capacity next year.
 - We believe a new fab of max capacity at 100k wpm is being built in Shanghai now, and is due to begin production early next year. There is no mention of this fab in any company filings and there is skepticism around its existence or whether it is controlled by CXMT. Nonetheless our checks from multiple sources suggest this fab exists and likely is being equipped. And it is likely closely related to CXMT's HBM production, which is why it is sensitive & may have been purposely hidden from its disclosures. For now we assume this fab is in progress and will begin to ramp in 1H27 and reach full capacity in 2028. And we also believe a second fab may be built in Shanghai and will add more capacity in 2028 additionally.
 - [Reuters reported](#) that CXMT was discussing with Beijing municipal government on securing land for building another fab in Beijing. We believe this very likely will go ahead and expect the fab to be completed and enter production sometime in 2028.
 - Apart from Shanghai & Beijing, CXMT's Hefei campus also has land for a 3rd fab. We believe construction is underway and production likely can begin in 2028 or even earlier.
- Overall we model CXMT's wafer capacity to reach 380K wpm at 4Q27 and 530K wpm by 4Q28, which is ~75% increase from now ([Exhibit 1](#)). This would make CXMT the third-largest DRAM maker globally in terms of wafer capacity. It would amount to 19% of the global DRAM wafer capacity and would also put CXMT ahead of Micron in terms of wafer capacity ([Exhibit 2](#)).
- On the other hand, we also understand that CXMT's technology is lagging. Its production yield may be lower. Both could make its bit density (number of bits per wafer) lower than rivals'. We also need to consider price separately as CXMT may have different like-for-like prices & a different mix. As the result, we highlight that looking at sheer wafer capacity share likely inflates CXMT's actual market presence, and will elaborate these considerations & dimension CXMT's bit and revenue share more precisely below.

EXHIBIT 1: We project CXMT's wafer capacity to rise by ~75% from now to 4Q28.

Source: TrendForce, Bernstein estimates and analysis

EXHIBIT 2: We project CXMT's wafer capacity share may surpass Micron in 2028, reaching nearly 20% globally.

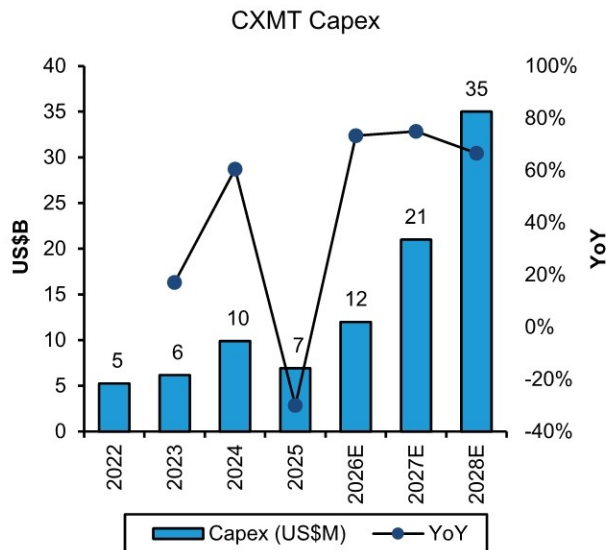
Source: TrendForce, Bernstein estimates and analysis

We model CXMT to spend more & more capex to support the expansion.

- CXMT's capex grew very quickly from 2022 to 2024, but the growth paused in 2025 probably because existing fab space was nearly all filled or CXMT anticipated tightening export restriction & bought equipment before actual needs. Going forward, to support the aggressive capacity expansion plan, we expect capex to grow rapidly again. We model its 2026 capex surging by over 70% YoY to reach US\$12B, and then continuing to rise at ~70% per year in the following two years, to eventually US\$35B in 2028, as the company tries to help China become more self-reliant on DRAM ([Exhibit 3](#)).
- In the past, CXMT's capex was multiple times higher than its revenue and CXMT had to rely on funding mainly from government at both central and local levels. Now with the steep memory price increase, CXMT will have no issue financing its expansion through operating cash flow ([Exhibit 4](#)). We think if it were not for physical limitations such as sourcing land,

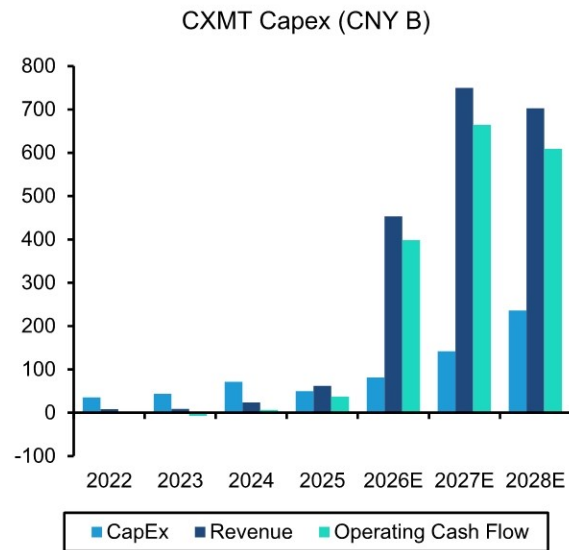
constructing fabs, recruiting & training employees, tuning equipment to make it production-worthy, etc. CXMT might be even more aggressive with capex and capacity expansion.

EXHIBIT 3: With strong cash position, we expect CXMT will increase capex spending quickly for further capacity expansion.



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 4: With strong operating cash flow starting this year, CXMT should have no problem financing its capex spending in the next few years.



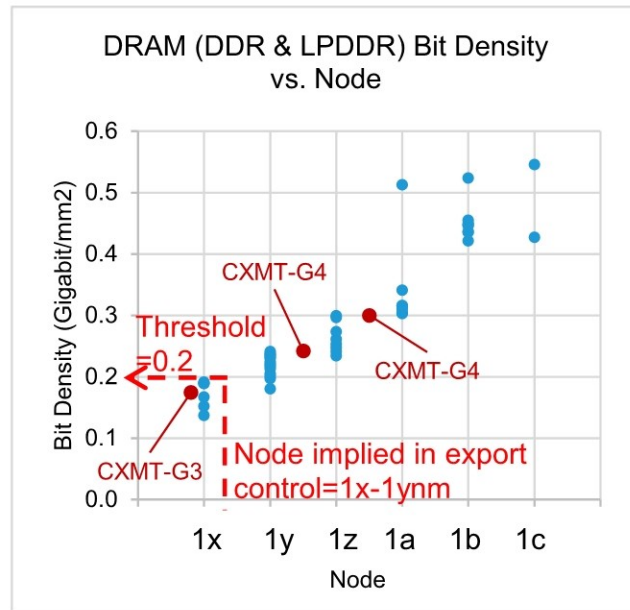
Source: Company reports, Bernstein estimates and analysis

GOOD ENOUGH TECHNOLOGY FOR CONVENTIONAL DRAM

CXMT is c. 3-4 years behind in technologies but is still capable of addressing most of DRAM market.

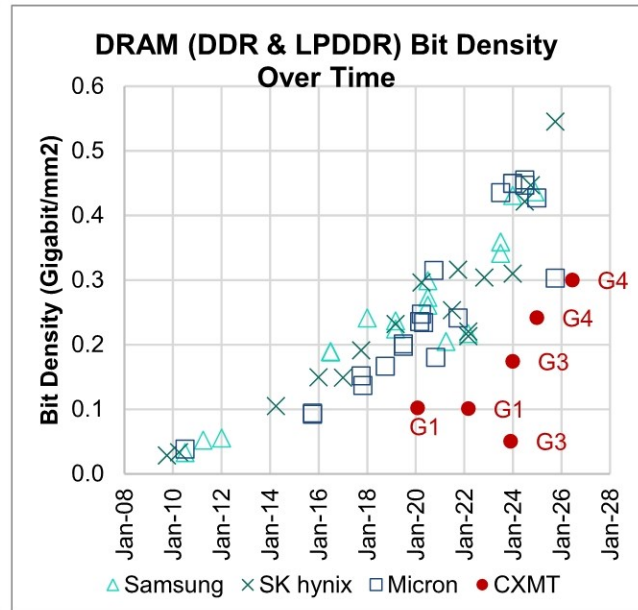
- As suppliers name their nodes differently, and CXMT further defines its technology by generations (e.g. G1, G2, etc., instead of 1nm, 1bnm, etc.), we hence look at bit density found in teardown reports to have a fair comparison on their technologies. As shown in [Exhibit 5](#), CXMT's latest sample on G4 has a density similar to 1z-1nm of others, implying that CXMT is c. 3-4 years behind the incumbent DRAM suppliers whose latest technology is 1c/1gamma nm ([Exhibit 6](#), [Exhibit 7](#)). The incumbents may advance to 1d/1delta nm in 2028, but we expect CXMT to shift to G5 in 2H27 to keep the gap about the same ([Exhibit 8](#)).
- The lagging technology however does not prevent CXMT from competing effectively in most of the DRAM market. Its DDR5 products likely have slower speed, but [recent reports finding CXMT breaking 9000 MT/s mark and narrowing the gap vs. rivals](#). [Xiaomi recently also confirmed to adopt LPDDR6 from CXMT](#), making CXMT one of the earlier suppliers mass-producing LPDDR6. Our checks also indicate that CXMT's price is similar to others', implying the share gain wasn't a result of heavy price discounts. Moreover, CXMT's disclosures show that CXMT has already derived 40% of its revenue from the relatively high-end DDR5 in 1H25 & 27% from server segment in 2025, respectively ([Exhibit 9](#), [Exhibit 10](#)). [Exhibit 11](#) shows the revenue composition of the global DRAM market. Compared to that, CXMT is clearly short on HBM, but the exposure to high-end applications such as server isn't much lower than the overseas leaders'.
- Most importantly we're surprised that CXMT defied the headwind of lagging technology, smaller scale, and less capable equipment (as the result of the export control), and delivered rapid margin expansion ([Exhibit 15](#)). In fact as of 2Q26, CXMT's gross margin was as high as 88%, very similar to what the larger rivals achieved ([Exhibit 12](#)).
- Our checks with consumer customers suggest that CXMT's price is similar to others', but we wonder if the comparable gross margin is a result of policy forcing SOEs & government agencies to pay higher prices. Even if this is true, this "subsidy" likely won't go away & will continue supporting CXMT, as China's localization policy very likely will remain in place for the foreseeable future.

EXHIBIT 5: ... and also from bit density point of view, but CXMT's G4 has exceeded the threshold.



Source: TechInsights & Bernstein analysis

EXHIBIT 6: CXMT's latest node is G4 and its density is roughly 3-4 years behind global leaders.



Source: TechInsights and Bernstein analysis

EXHIBIT 7: CXMT's G4 is roughly equivalent to 1z or 1a nm from others, but others have advanced their technologies to 1c/1gamma nm.

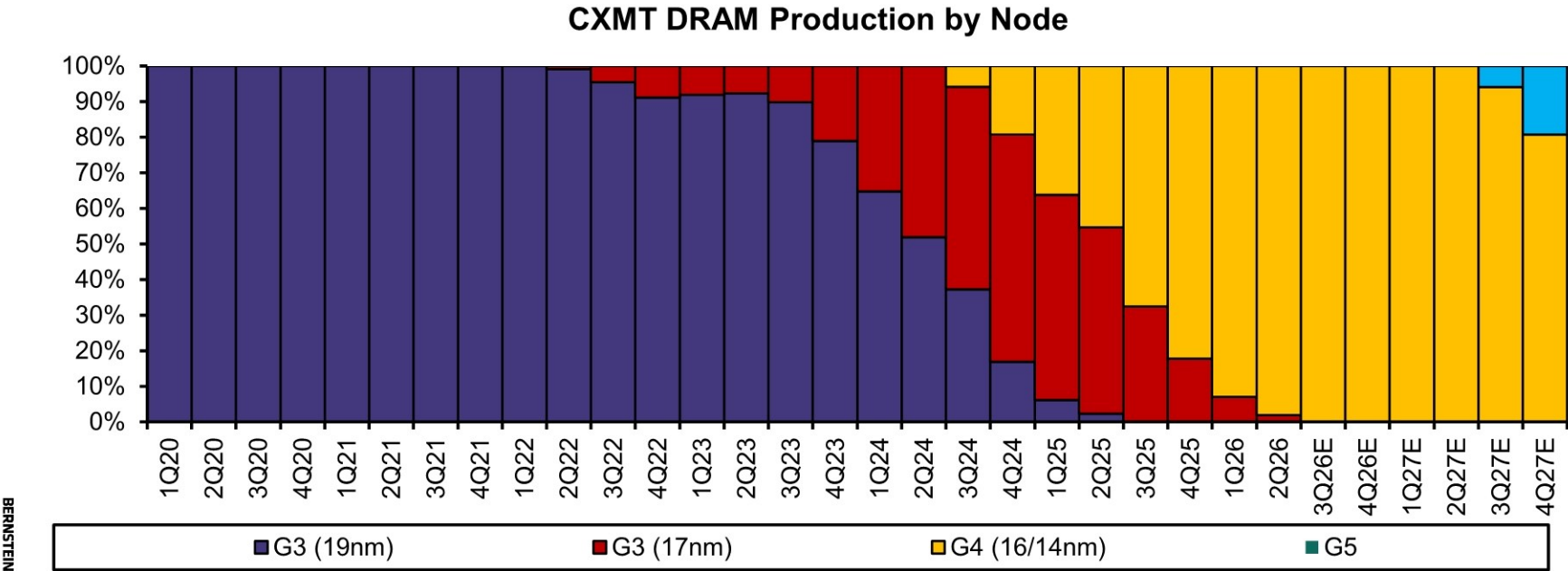
Updated on Sep 13, 2026

	2025 MS	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Samsung	34%	28 nm	25/23nm	20nm	18 nm (1x)	17 nm (1y)	15/16 nm (1z)*	13/14 nm (1a)	12 nm (1b)	1c nm	1d nm						
SK hynix	36%	29 nm	25 nm	21 nm	18 nm (1x)	17 nm (1y)	15/16 nm (1z)	14 nm (1a)*	12 nm (1b)	1c nm	1d nm						
Micron	22%	30 nm	25 nm	20 nm	17 nm (1x)	17nm (1y)	15/16 nm (1z)	14 nm (1a)	13 nm (1β)	1γ nm*	1δ nm						
Nanya	1%	30 nm	2x nm	1A nm	1B nm	1C nm*											
CXMT	6%						G1 (2y nm)	G3 (1x-1γnm)	G4 (1z-1anm)	G5							

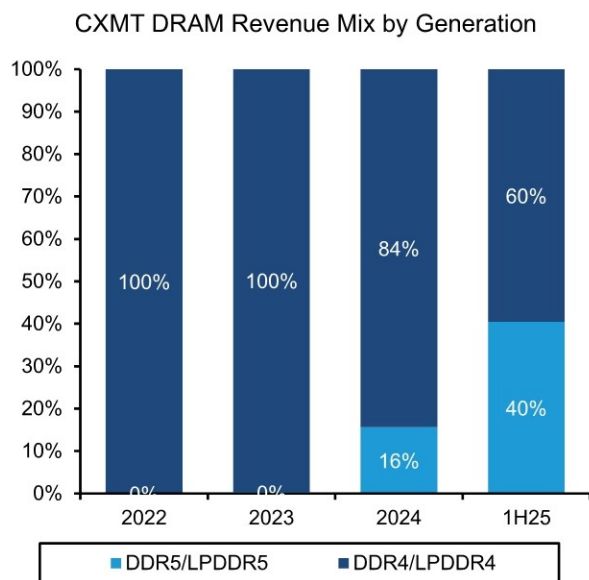
*: Beginning of EUV adoption in mass production; Samsung adopted EUV in 1x & SK hynix in 1y very limitedly

Company release, and Bernstein estimates and analysis.

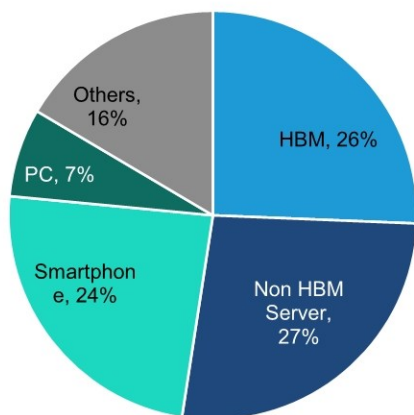
EXHIBIT 8: We believe CXMT will begin the migration to G5 in 2027.



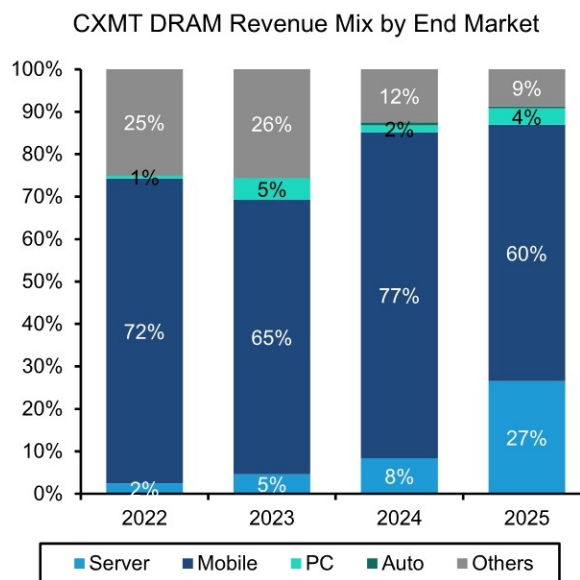
Source: TrendForce, Bernstein estimates and analysis

EXHIBIT 9: 40% of CXMT's DRAM revenue came from DDR5/LPDDR5 as of 1H25.

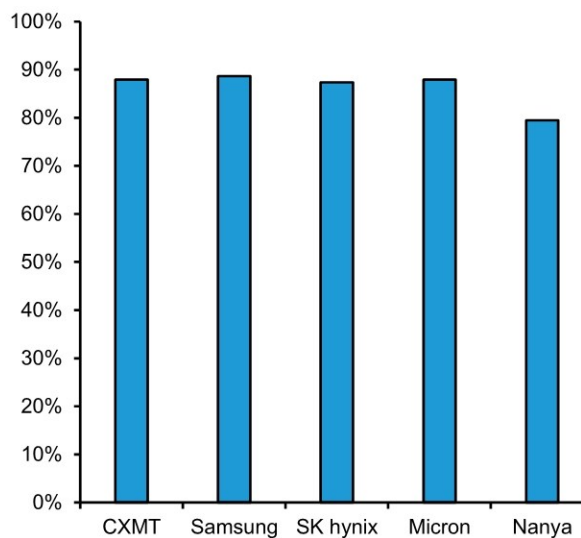
Source: Company reports and Bernstein analysis

EXHIBIT 11: Globally, server DRAM is 25-30% of DRAM market revenue in 2025.**Global DRAM Revenue by End Market (2025)**

Source: Company reports, TrendForce and Bernstein analysis

EXHIBIT 10: 27% of DRAM revenue at CXMT was from servers in 2025.

Source: Company report and Bernstein analysis

EXHIBIT 12: As of 2QCY26, CXMT DRAM gross margin already reached a level similar to global incumbent players'.**2QCY26 DRAM Gross Margin**

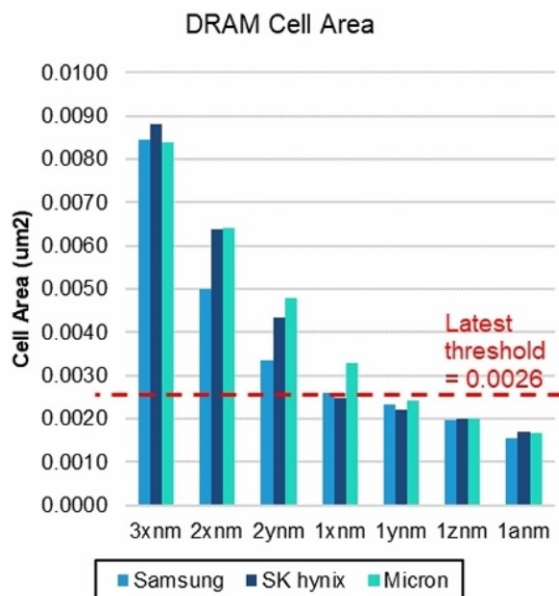
Source: Company reports and Bernstein analysis

CAN EXPORT CONTROL KEEP CXMT FROM GROWING & FROM AI (I.E. HBM)?

Export restriction imposed by the US is a notable risk to the capacity & capex projection above.

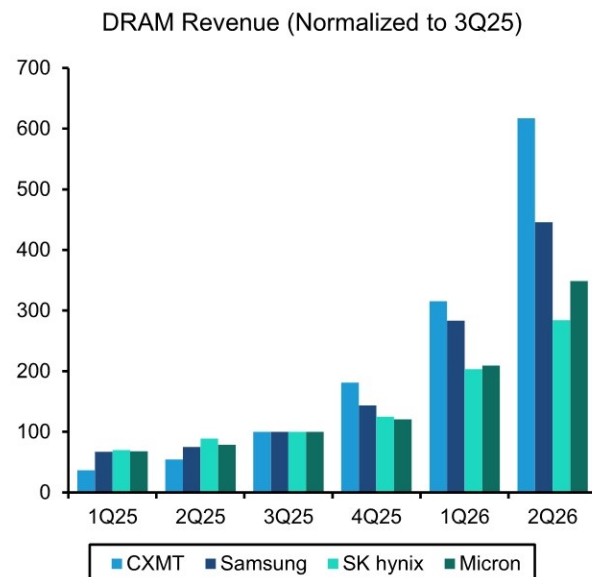
- The current restriction was announced by the US in Jan 2025. As analyzed in [this report](#), the restriction keeps China, CXMT included, from importing equipment capable of producing DRAM with “cell area $< 0.0026\mu\text{m}^2$ ” or “bit density $> 0.20\text{Gb}/\text{mm}^2$ ”, approximately equal to 18nm (i.e. between 1x & 1y) DRAM ([Exhibit 13](#), [Exhibit 5](#)). CXMT’s ability to produce HBM is also limited as the restriction additionally controls equipment capable of producing “ > 3000 TSVs per die”. Finally, we note this restriction applies to non-US tools, and some high-end DUV (& certainly EUV too) systems from ASML are hence unavailable to CXMT.
- However after the restriction, CXMT surprisingly continued to expand & improve. Looking at [Exhibit 5](#) closely, investors should note that CXMT’s current generation G4 has already exceeded the boundary set by the export control. And its expansion and improvement do not seem to be simply a result of riding the current cyclical wave, as they also happened in 1H25 before the cycle started. It could be attributable to the “latent” capacity released from the imported equipment that CXMT stocked up beforehand, but if so the “latent” benefit is still helping CXMT even 1.5 years after the restriction turned effective, as in 2H25 & 1H26 CXMT also enjoyed more improvements than rivals ([Exhibit 14-Exhibit 15](#)). We hence make an important assumption & project CXMT can continue expanding effectively from now to 2028 despite the export restriction.
- That said, the benefit of the stockpiled equipment will eventually fade, and it will be a downside risk to our projection if CXMT can’t find good enough Chinese alternative in time. The advancement of CXMT’s technology, for example from G4 to G5, may also make it more difficult for local equipment suppliers. CXMT may experience yield difficulty and its expansion may have to slow down. A further downside to our projected expansion is that the US for example does not feel the dependence on rare earth from China any longer & decides to further tighten the restriction on CXMT. We will discuss how this would affect CXMT’s HBM progress & its long-term technology roadmap later in this report.

EXHIBIT 13: Current threshold of export control for DRAM production equipment is between 1x & 1ynm from cell area perspective ...,



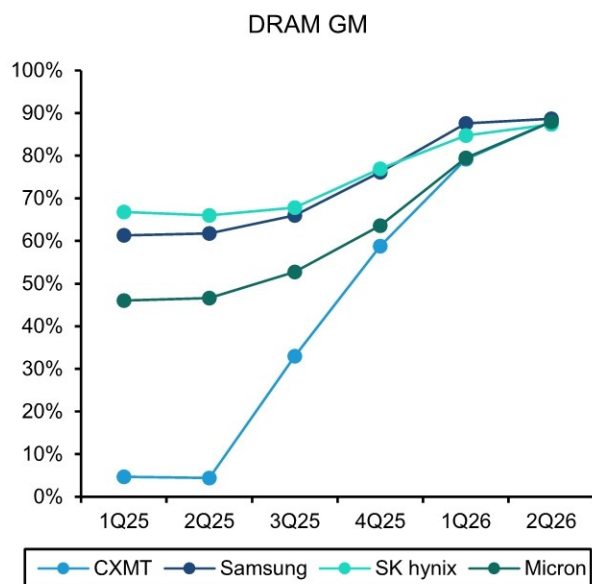
Source: TechInsights and Bernstein analysis

EXHIBIT 14: After the export control announced in early 2025, CXMT continued to grow, and the growth existed even before the current cycle started.



Source: Company reports and Bernstein analysis

EXHIBIT 15: 1.5 years after the restriction, CXMT continued to improve. Its profitability has been comparable to the incumbents for 2 quarters.

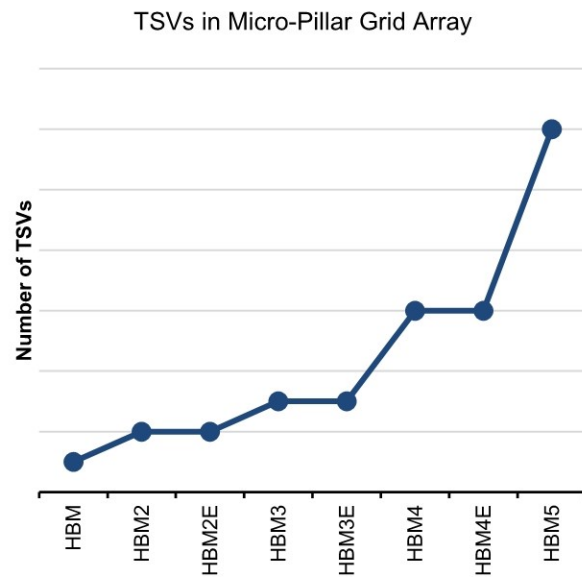


Source: Company reports and Bernstein analysis

HBM is technologically difficult for CXMT, but we expect China to compromise with its own versions of HBM & CXMT's profitability will be protected by policy.

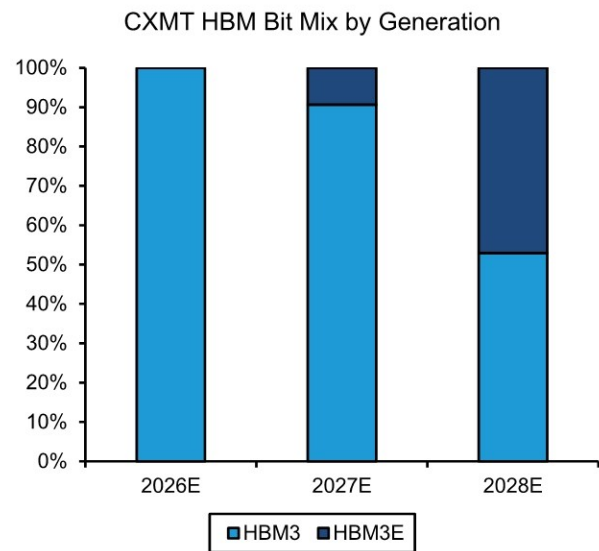
- As HBM demands better performance and is difficult to yield, it will present a notable technology hurdle to CXMT. The hurdle will be in both front-end wafer fabrication & back-end packaging and very likely will make CXMT's HBM yield far lower than others'. CXMT likely has limited experience in the production steps unique to HBM, e.g. TSV & wafer/die bonding, and will find them challenging for example. Finally, the current export restriction bans China from importing equipment capable of ">3000 TSVs per die" but [SK hynix puts >5600 TSVs per die in HBM3](#) and [>20000 in HBM4](#). With technology complexity demands even denser TSVs in the future ([Exhibit 16](#)), the export control is becoming increasingly restrictive for CXMT. Finally, as Samsung adopts 4nm logic process in its HBM4 base die, advanced logic nodes are necessary to make base die faster enough for the future generations of HBM. Given China's struggle in advanced logic manufacturing, CXMT will find base die a difficulty in the future too. All these will make CXMT's HBM yield low and cost high.
- However, the restriction set by both the US & China on importing GPUs has effectively divided HBM demand into two separate markets. Though Chinese AI labs & CSPs focus more on AI applications instead of frontier models, and have been innovating to reduce the use of memory including HBM, HBM is still necessary for many aspects of AI developments & deployments. They will find imports insufficient to fulfill their HBM demand, and will have no choice but to source HBM from CXMT. These customers thus will have to share the high cost, & CXMT's HBM profitability will thus be protected. Additionally, we also expect China to develop its own flavors of HBM, and "HiBL 1.0" & "HiZQ 2.0" announced by Huawei are an example. They will be paired with Ascend 950PR for pre-fill, etc. with lower cost & Ascend 950DT for decoding & training with higher cost, respectively ([Exhibit 18](#)). Given China's need to localize HBM, but also CXMT's yield difficulties with HBM, we expect such a compromise to extend to other Chinese AI accelerators too.
- We hence project CXMT to progress ahead from HBM3 (or something equivalent) this year to HBM3E in 2027 ([Exhibit 17](#)) but stay c. 3-4 years behind rivals ([Exhibit 19](#)). HBM will consume more capacity at CXMT gradually ([Exhibit 20](#)), and is projected to account for 9% of CXMT's revenue in 2028. We also model CXMT to get higher HBM price ([Exhibit 22](#)) to mitigate the yield & cost difficulty so that the gross margin of HBM, though not as good as that of conventional DRAM, will stay satisfactory in our forecast horizon ([Exhibit 23](#)).

EXHIBIT 16: Higher technology complexity and higher TSV density will make the existing export control increasingly difficult for CXMT.



Source: Samsung and Bernstein analysis

EXHIBIT 17: We model CXMT's HBM production to begin with HBM3 and start to shift to HBM3E next year.



Source: Bernstein estimates and analysis

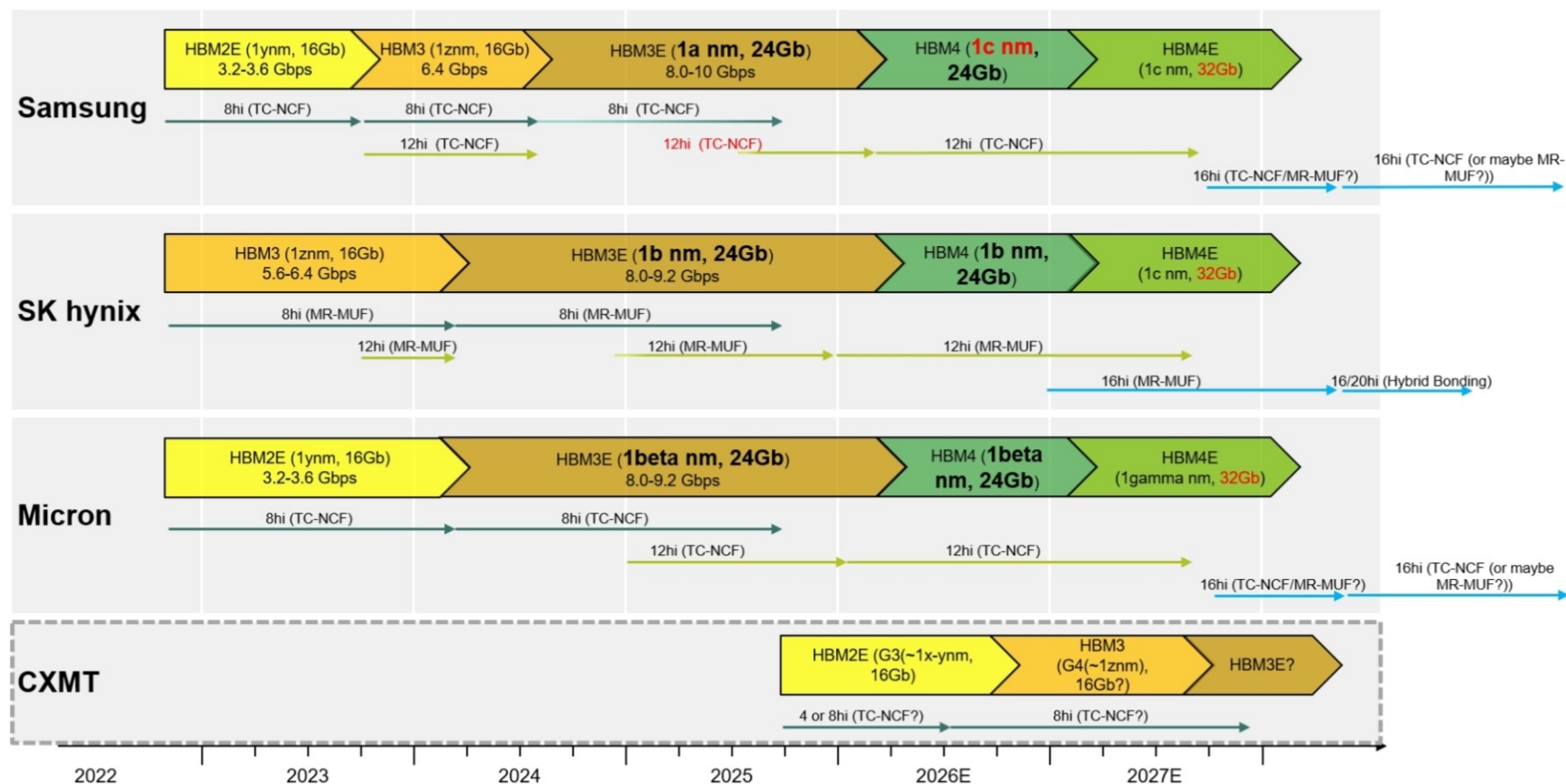
EXHIBIT 18: Huawei's Ascend 950 AI accelerators come equipped with its own design of HBM, named HiBL and HiZQ.

Chip	Launch	Usage	Interconnect bandwidth	Compute power	HBM	Memory capacity	Memory bandwidth
Ascend 950PR	2026 Q1	Prefill and Recommendation	2 TB/s	1 PFLOPS FP8, 2 PFLOPS FP4	HiBL 1.0	128 GB	1.6 TB/s
Ascend 950DT	2026 Q4	Decode and Training	2 TB/s	1 PFLOPS FP8, 2 PFLOPS FP4	HiZQ 2.0	144 GB	4 TB/s

Source: Huawei and Bernstein analysis

EXHIBIT 19: CXMT's HBM technology likely remains c. 3-4 years behind the incumbents'

HBM Supplier Technology Roadmap

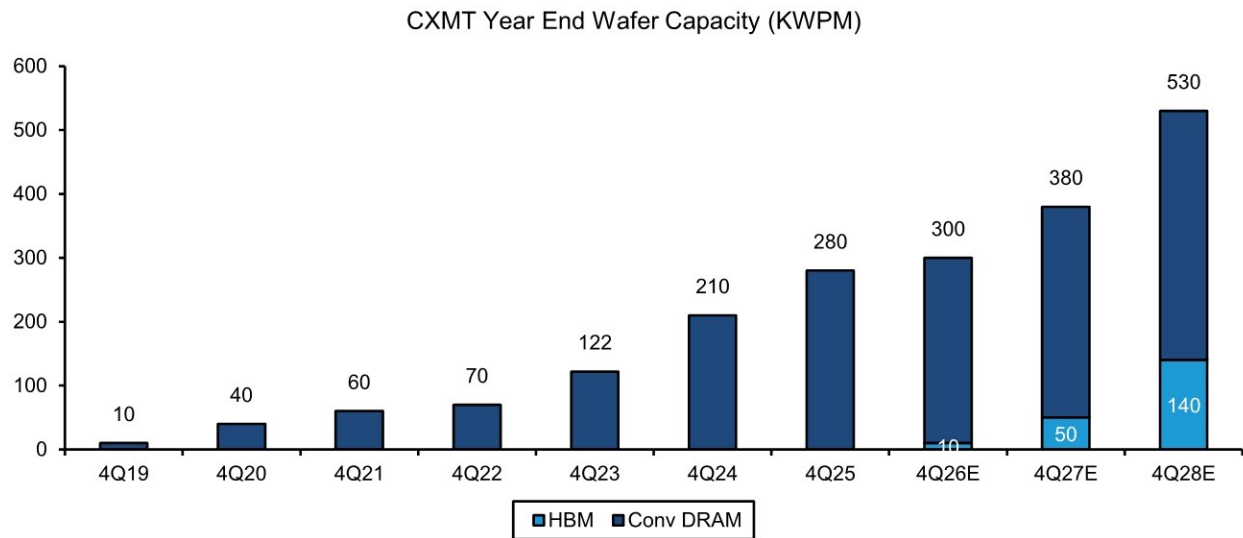


Note

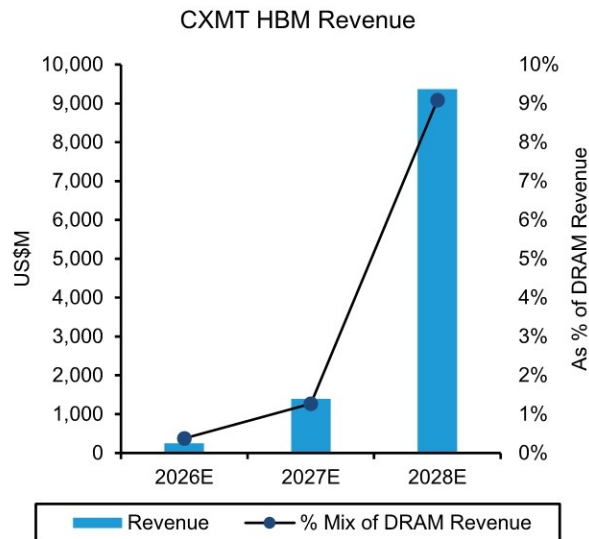
- TC-NCF: thermal compression bonding with non-conducting film
- MR-MUF: mass reflow-molded underfill

Last Revision: Sep 7, 2026

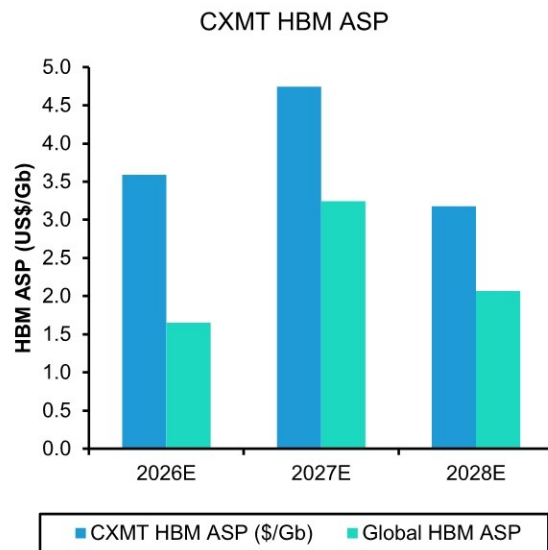
Source: TrendForce, Bernstein estimates and analysis

EXHIBIT 20: **We project HBM to take up more wafer capacity at CXMT going forward.**

Source: TrendForce, Bernstein estimates and analysis

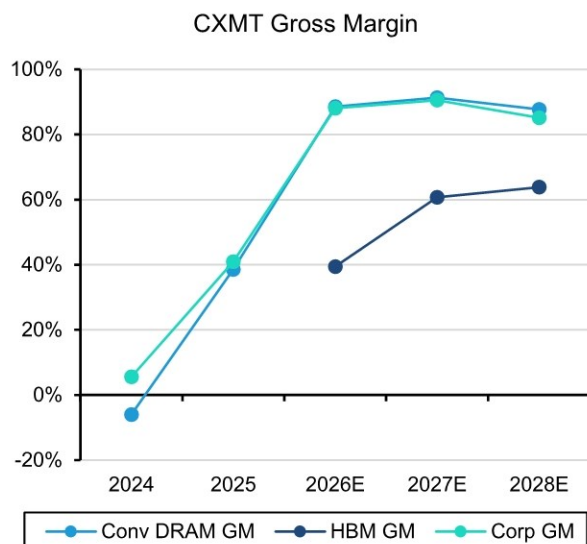
EXHIBIT 21: **We model CXMT's HBM revenue to rise to 9% of its total DRAM revenue in 2028.**

Source: Bernstein estimates and analysis

EXHIBIT 22: **We expect CXMT's HBM will command a price premium over global suppliers' products to compensate for yield and cost difficulty.**

Source: Bernstein estimates and analysis

EXHIBIT 23: We expect localization policy will make CXMT's HBM revenue earn satisfactory gross margin.



Source: Company reports, Bernstein estimates and analysis

DUV may be enough to sustain CXMT's competitiveness for 3-5 years. Moreover, whether other equipment emerges as a bottleneck remains to be seen.

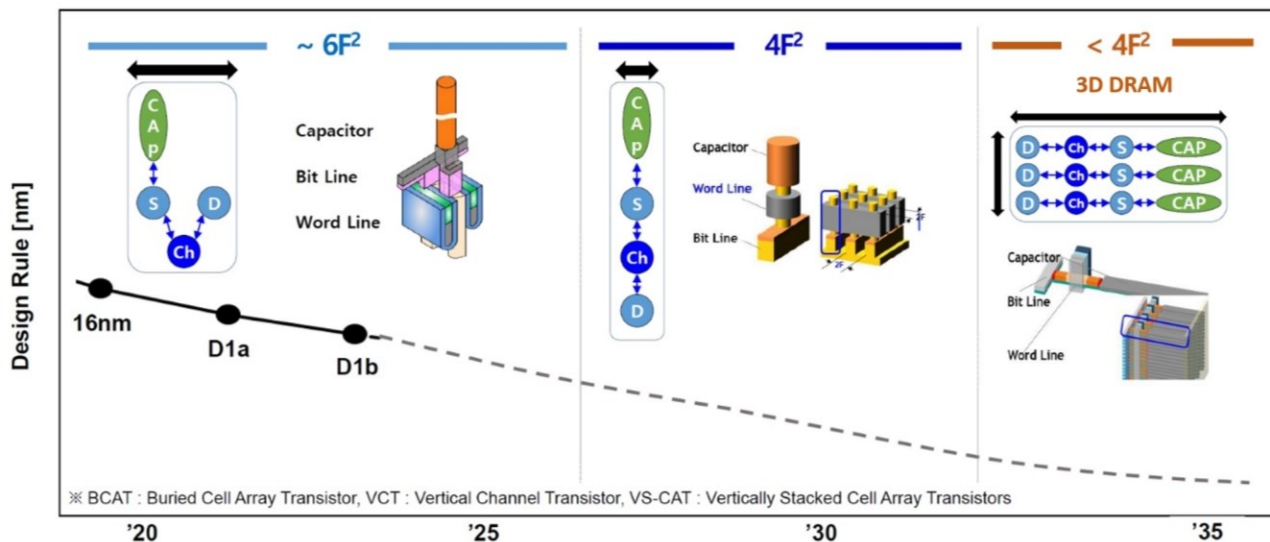
- As Micron does not adopt EUV until 1 gamma nm, it suggests that DUV can effectively produce DRAM up to 1b/1beta nm. For CXMT, we expect the company to stretch the reach of DUV further, possibly to 1c/gamma nm or even one more node. Such a stretch will result in yield difficulties and deteriorating economy for CXMT. However, this burden will be shared by Chinese government and domestic supply chain, and accordingly it is conceivable that even constrained at DUV CXMT can stay competitive for 2-3 generations (from 1z-1anm now to possibly 1c-1dnm), or equivalently for another 3-5 years.
- Furthermore, foreign non-lithographic equipment is also restricted under the existing export control. As Chinese semicap suppliers generally offer more capable alternative to the imported equipment in these areas, we assume this is less of an issue for CXMT but will continue to monitor if CXMT's expansion & technology development can proceed as planned.
- Notably though CXMT is restricted on advanced equipment, the export control actually allows most of these tools to be sold to other fabs in China & certainly in other countries. This makes enforcement difficult, and somehow subject to the will and resources that US government has, as a "grey second-hand market" may emerge and some restricted equipment or spare parts may eventually find their ways to CXMT.
- On the flip side, the discussion above also implies that for the investors of the incumbent DRAM suppliers, both cycle sustainability & the threat from CXMT are important investment considerations.

Lack of EUV limits CXMT's long-term runway, but CXMT is trying to overcome that with 4F² and 3D DRAM. Neither of them is easy though.

- CXMT is well aware that lack of EUV is a long-term issue, and has been devoting R&D resources to technologies that promise to circumvent the need of EUV in DRAM. 4F² differs from the 6F² structure employed in DRAM currently and can shrink DRAM cell size without better lithography ([Exhibit 24-Exhibit 25](#)). 4F² is however a one-time fix, as after the 4F² adoption further shrink will still rely on EUV. 3D DRAM builds more & more DRAM cells vertically in a wafer. It differs from the 2D structure currently and will fundamentally remove the need of EUV. Further density improvement will be from building more cells vertically and will not require EUV, as we saw when NAND transitioned from 2D to 3D. Our recent [AI memory primer](#) discussed 4F² and 3D DRAM more, as well as other important memory technologies for AI.

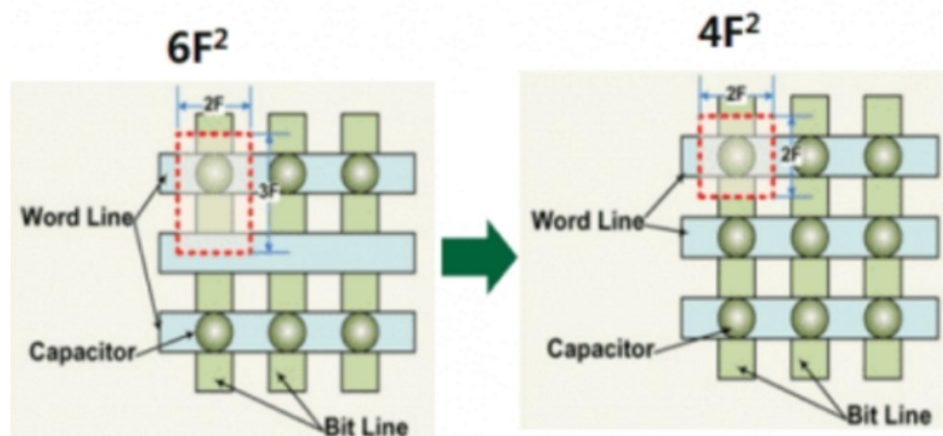
- That said rivals have been exploring both $4F^2$ and 3D DRAM for years too. Their universal decision to use EUV over these two suggests that $4F^2$ & 3D DRAM have technological difficulties & uncertain economy and EUV is thus a better choice. We thus for now assume neither of the two innovations will be commercially competitive and CXMT will become increasingly less competitive 3-5 years from now.

EXHIBIT 24: **$6F^2$ is the prevailing structure of DRAM cell, but the industry, including CXMT, is trying to advance to $4F^2$ structure & to 3D DRAM.**



Source: Samsung, Bernstein analysis

EXHIBIT 25: **A top view of DRAM cell arrays to illustrate the change from $6F^2$ to $4F^2$. By moving peripheral circuits below capacitor in the same wafer, or even to a different wafer, $4F^2$ architecture reduces the area of a DRAM cell from $6F^2$ to $4F^2$.**



Source: Samsung

COMPETITIVE ONLY IN CHINA & IN LOW-END SEGMENTS?

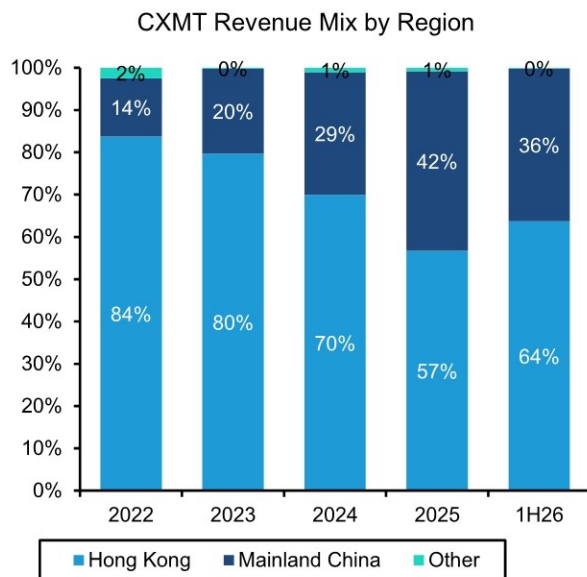
CXMT has been selling its products mostly to domestic customers, but we expand CXMT to expand its customer base to non-Chinese customers over time despite geopolitical concerns.

- CXMT reports that most of its revenue has been from Hong Kong, with the rest from Mainland China ([Exhibit 26](#)). However CXMT indicated this is because majority of CXMT's customers choose to transact in USD, and settlement is considerably more convenient in Hong Kong. Hence, we assume many Chinese customers set up entities in Hong Kong for this reason, as well as for trading, distribution, etc., and assume CXMT generates most of its revenue from Chinese customers. In 2025, revenue mix directly from Mainland China increased, mainly because more customers requested to settle in RMB.
- Some argue geopolitical concerns will make CXMT unable to address overseas market. However, Chinese OEMs have been exporting CXMT's memory to other countries, when they export their PC, smartphones, etc. overseas. [Lenovo exporting PC with YMTC's NAND-based SSD in Germany is an example](#). [Chinese AI models being used overseas is another](#), at least the training part even when they are open sourced & foreign users deploy these models on servers outside of China.
- Our checks also find many non-Chinese OEMs & module makers are OK to install CXMT's DRAM (& YMTC's NAND too) in products sold globally except in the US. [Reportedly Apple is trying Chinese memory too](#). All these make us conclude that CXMT may not be able to penetrate customers that are geopolitical sensitive (e.g. foreign AI model labs, hyperscalers, government agencies), but CXMT will be able to address the rest of the overseas DRAM market.

Application-wise, CXMT's DRAM has been used in basically all applications, with quality sufficient to derive 25-30% of its total revenue from servers in 2025.

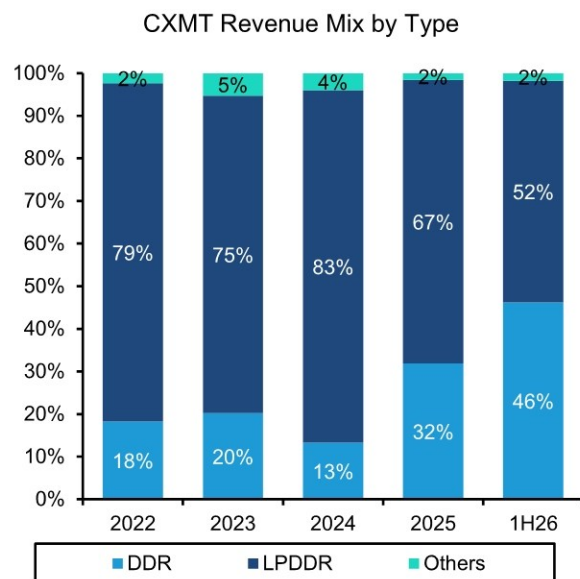
- CXMT started with LPDDR for mobile applications ([Exhibit 27](#), [Exhibit 10](#)), but entered DDR and other applications soon after. In 2025, CXMT derived over 30% of its revenue from DDR, and 25-30% from servers. Servers contributing significant revenue is particularly important, as they require higher quality and the contribution attests CXMT's product quality is already sufficient to satisfy this requirement.

EXHIBIT 26: CXMT's revenue almost entirely comes from mainland China and Hong Kong



Source: Company reports and Bernstein analysis

EXHIBIT 27: Majority of CXMT's revenue has been from mobile DRAM though DDR's mix has been increasing rapidly too.



Source: Company reports and Bernstein analysis

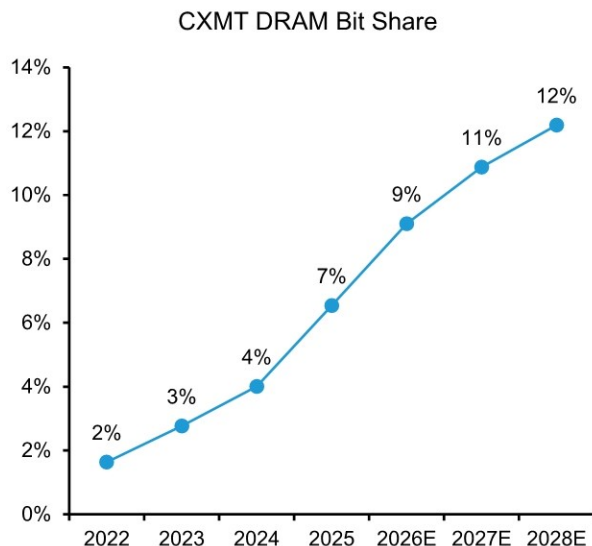
With lagging technology, CXMT's bit share will be lower than its wafer capacity share but is still expected to rise from 8% in 2026 to 13% in 2028.

- While we expect CXMT will have more wafer capacity than Micron in late 2028, we also highlight that CXMT's technology is lagging and won't generate as many bits per wafer as rivals'. CXMT's bit share accordingly will be lower than its wafer share, but is still modeled to rise from c. 9% this year to 12% in 2028 ([Exhibit 28](#)).
- We further highlight this accounts for the difficulty in HBM and thus CXMT's lower share in HBM vs. conventional DRAM. Should we focus on conventional DRAM only, CXMT's share will increase from c. 10% this year to 14% in 2028.

We project CXMT's revenue share similar to its bit share, as conventional DRAM remains the bulk of the DRAM market & its commodity nature will make CXMT's price similar to others'.

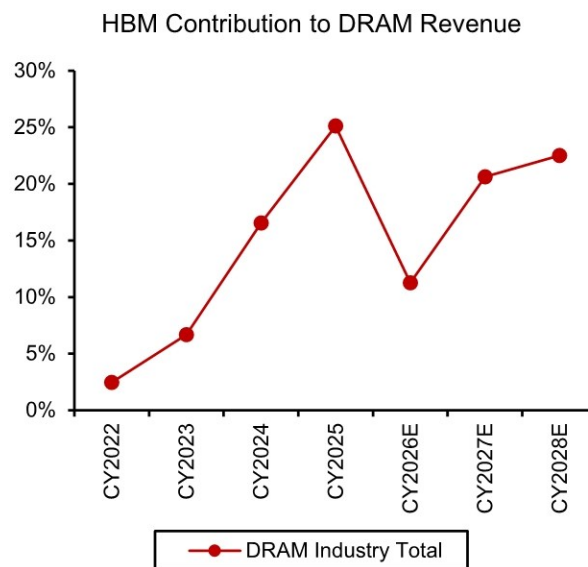
- As shown in [Exhibit 29](#), we forecast HBM's contribution will hover around c. 20% of the overall DRAM market & conventional DRAM will still represent most of the market in the next several years. It is because conventional DRAM enjoys a larger price hike than HBM in the near term. More importantly the structural shift from training to inference and agentic workload tilts the mix from HBM to conventional DRAM in the long run too.
- Export control divides HBM, & AI processor in general, into two separate markets & likely will make HBM's price/economy differ significantly in China vs. elsewhere. For conventional DRAM, though we recognize the policy of localization may help CXMT enjoy higher price at geopolitically-sensitive Chinese customers, most of CXMT's conventional DRAM is however sold to civilian applications or private sectors in which CXMT's price has to be competitive.
- Given that, and given conventional DRAM remains the bulk of the DRAM market, we model CXMT's corporate blended ASP similar to others'. This in turn will make CXMT's revenue share similar to its bit share. As shown in [Exhibit 30](#), we project the revenue share to reach 13% in 2028, up from 10% this year.

EXHIBIT 28: We model CXMT's share to rise steadily despite its lagging technology.

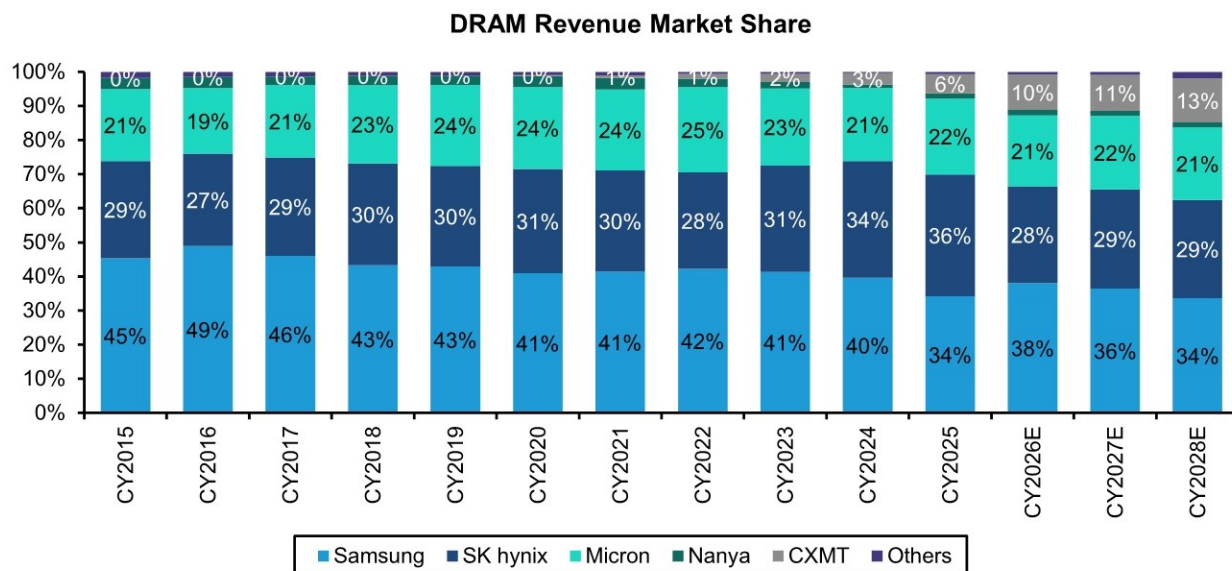


Source: Company reports, Bernstein estimates and analysis

EXHIBIT 29: We believe HBM will be minority & conventional DRAM will stay the majority of the industry total DRAM revenue in the next several years.



Source: TrendForce, company reports, Bernstein estimates and analysis

EXHIBIT 30: **We forecast CXMT's revenue market share to reach 13% in 2028.**

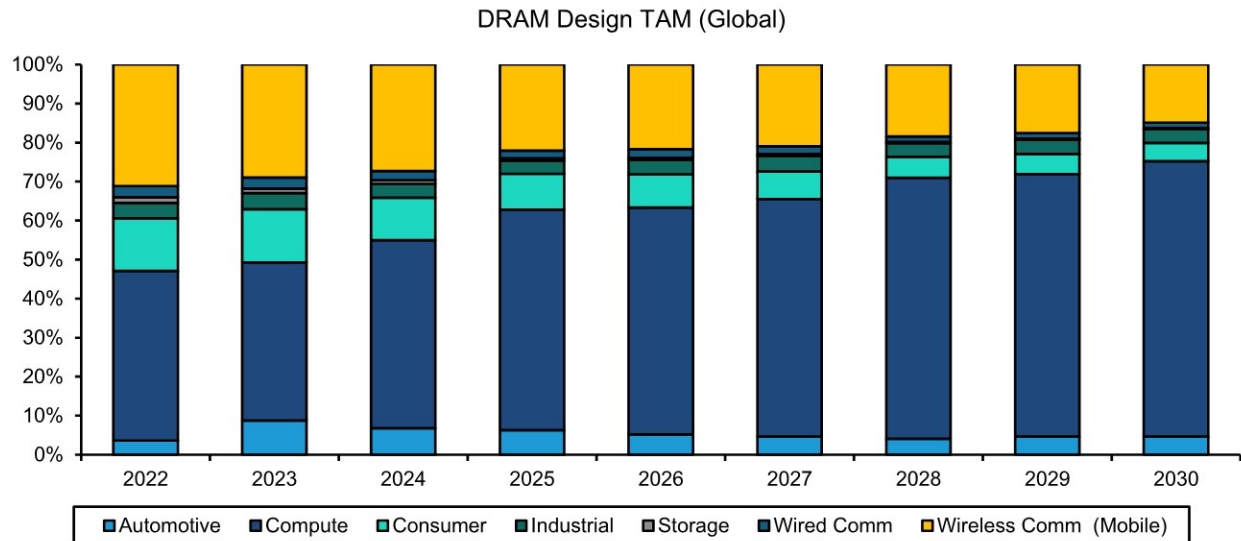
Source: Company reports, Bernstein estimates and analysis

CAN CHINA BE SELF-SUFFICIENT IN DRAM?

We estimate Chinese OEMs represent 20-25% of the global DRAM demand. Fulfilling only 40-45% of that now, CXMT still has ample growth headroom from domestic customers.

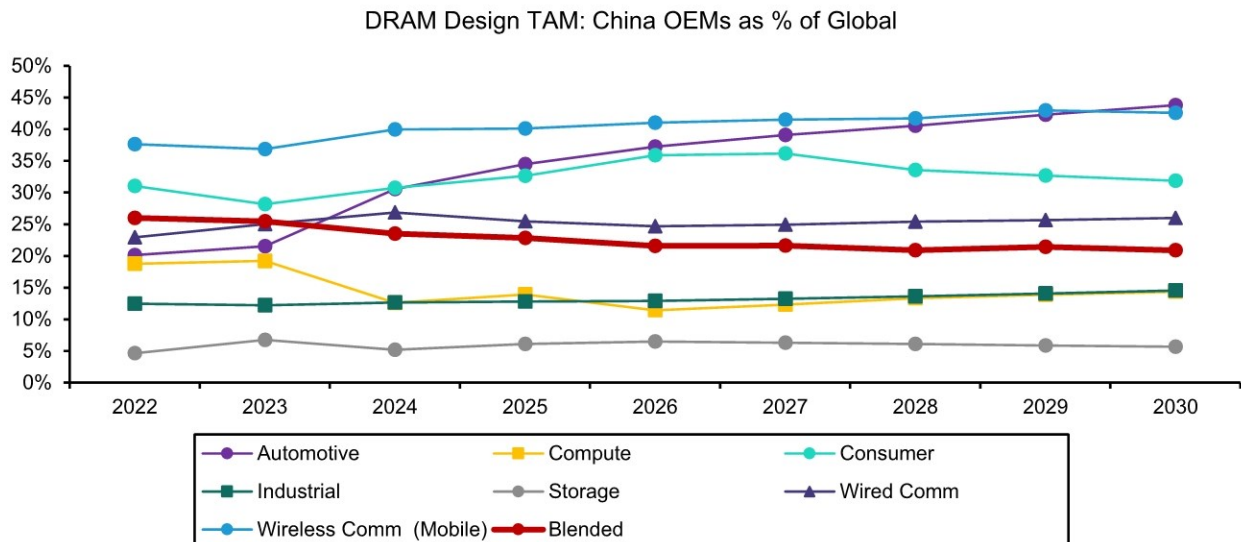
- As discussed above, Chinese OEMs installing CXMT's DRAM in their products & selling the products abroad will extend CXMT's TAM to other countries. Investors thus should not assume CXMT's TAM is limited to China's domestic consumption. We call the amount of DRAM procured by OEMs "design TAM" and will use that to estimate how much self-sufficiency CXMT helps China achieve in DRAM.
- Gartner estimates that in 2026, 58% of the global DRAM demand will be from compute applications (e.g. PC & server). Mobile applications (mostly smartphone) is the second-largest segment and will account for 22%. As more & more AI data centers are being built, Gartner forecasts the % contributed by compute will rise, to 67% in 2030 ([Exhibit 31](#)).
- Gartner further considers Chinese OEMs' share in the global design TAM for different applications. For example China's large presence in handset ecosystem explains why Chinese OEMs' share in the mobile design TAM is higher at 41%. The share in computing is lower, and will become even lower as other countries build more AI data centers than China. We then calculate a weighted average and estimate Chinese OEMs represent about 20-25% of the overall DRAM demand globally this year ([Exhibit 32](#)).
- Combining that with the assumption that all CXMT's revenue is from Chinese OEMs, we estimate CXMT enables China to achieve ~40-45% self-sufficiency in DRAM this year ([Exhibit 33](#)).
- We can then further conclude growth headroom is still ample and should be sufficient to absorb most of the rapid capacity expansion that CXMT is set to do in the next few years.

EXHIBIT 31: Driven by AI data centers compute application is expected to represent a bigger part of the global DRAM market in the next 5 years.



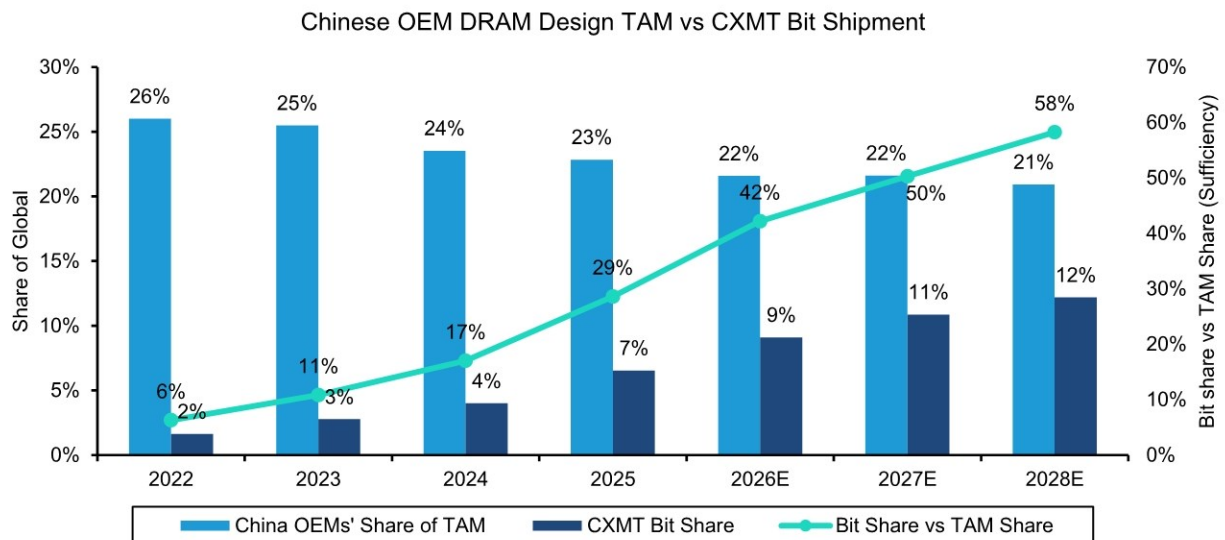
Source: Gartner and Bernstein analysis

EXHIBIT 32: China OEMs have a high share in DRAM demand in mobile end market. However with a much lower share in compute end market, their overall blended share is 20-25% of the global DRAM demand.



Source: Gartner and Bernstein analysis

EXHIBIT 33: Comparing with China OEMs' share of DRAM design TAM with CXMT's market share, we estimate CXMT enables China to achieve 42% self-sufficiency this year.

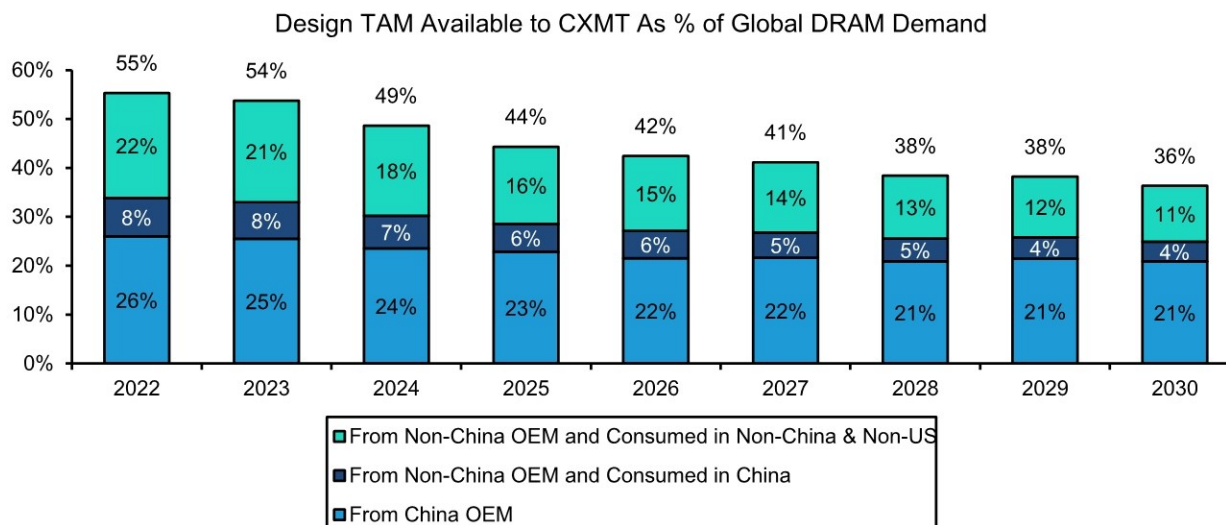


Source: Gartner, Company reports, Bernstein estimates and analysis

We believe consumer applications sold outside of the US can be addressed by CXMT too & should be part of its TAM. We estimate CXMT to capture only 21% of this “expanded TAM” this year, implying even bigger growth headroom.

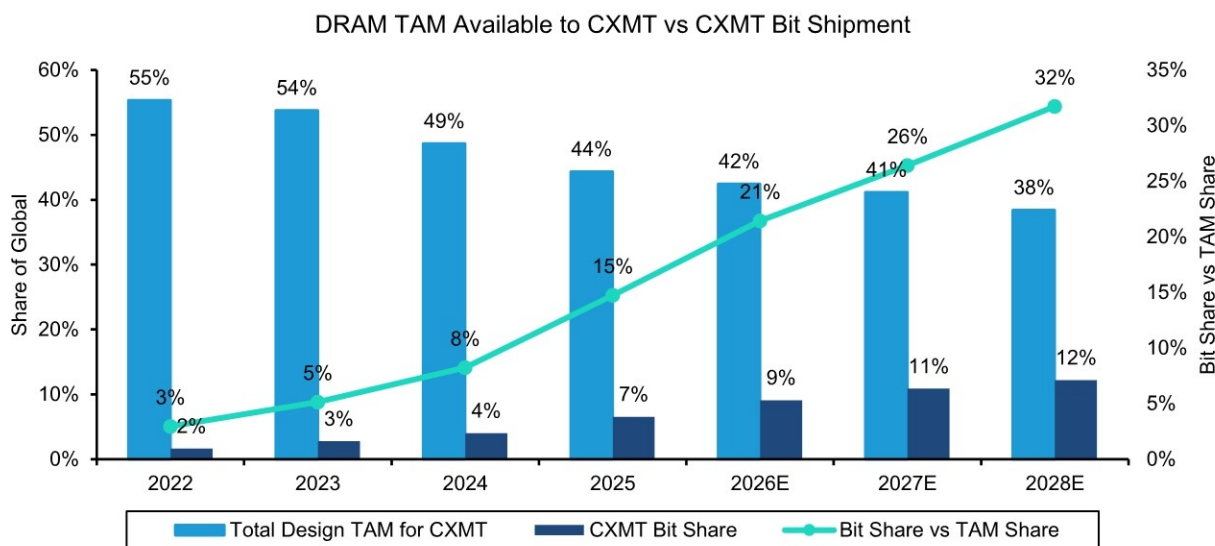
- As foreign OEMs may choose to adopt CXMT DRAM in the products sold in countries outside the US, we find the TAM addressable by CXMT can expand to 42% of the global DRAM demand this year. This “expanded TAM” assumes only PC, mobile and consumer applications, but excludes demand from server, auto, etc. on geopolitical concerns ([Exhibit 35](#)). Compared to this “expanded TAM”, CXMT’s share will be only about 21% this year, implying a growth headroom even bigger than what we discuss above ([Exhibit 35](#)).

EXHIBIT 34: If we also include foreign OEMs DRAM demand for consumer products sold in China and other non-US countries as the TAM available to CXMT, the TAM can be 42% of the global DRAM demand this year & 36% in 2030, based on Gartner's projection.



Source: Gartner and Bernstein analysis

EXHIBIT 35: As CXMT's global share is only at 9% in 2026, it implies CXMT's penetration of the "expanded TAM" is just 21% this year.



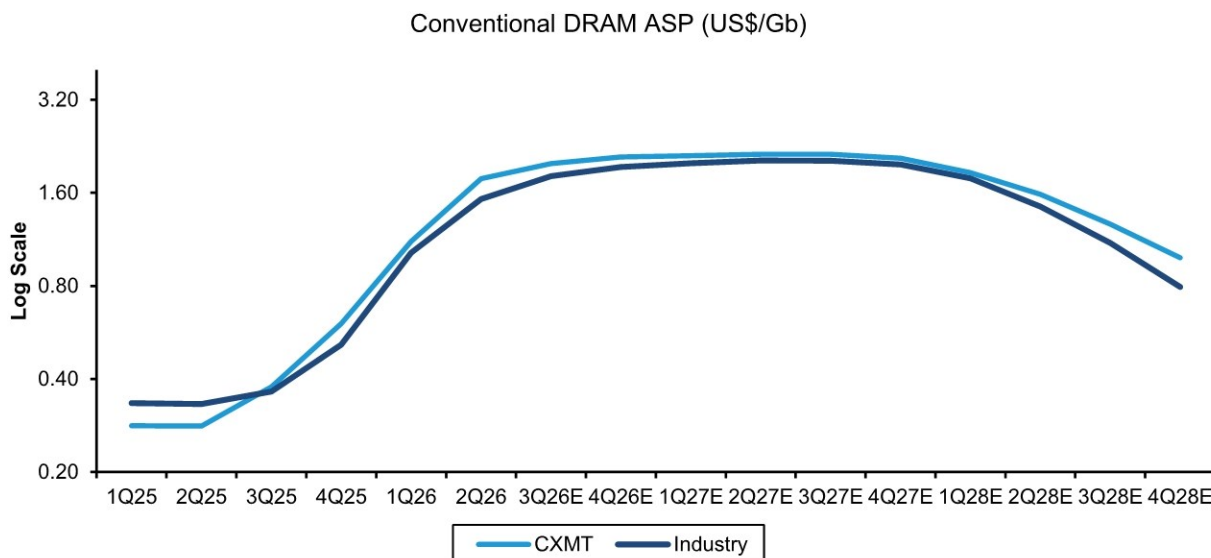
Source: Company reports, Gartner, Bernstein estimates and analysis

FINANCIAL PROJECTION COMBINING BOTH CYCLICAL & IDIOSYNCRATIC PERSPECTIVES

Robust price through 2027, but policy & catching-up HBM should make CXMT more resilient when the market “normalizes” in 2028.

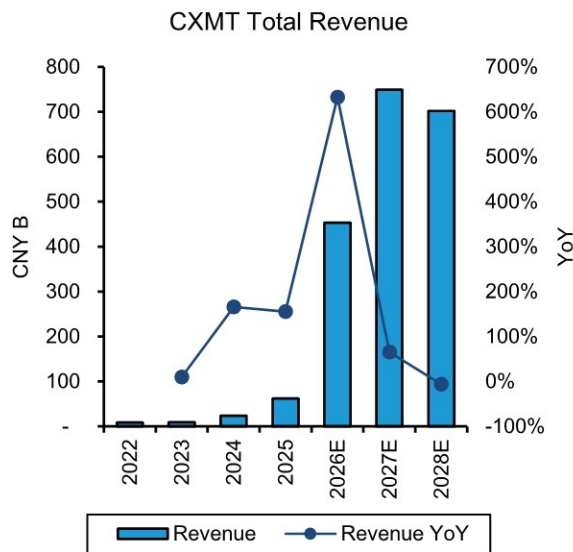
- Based on its financial results, we believe CXMT enjoyed a steeper price increase from 3Q25 than the global incumbents. We don't think there is material price disparity between CXMT and the incumbents, and perhaps CXMT may enjoy some price premium by selling more through distribution channels, where prices likely are higher than mainstream contract prices. Going forward, we expect some further price increase for both the industry and CXMT. And we expect DRAM price to stay elevated or inch higher for most of 2027 as DRAM shortage persists.
- Into 2028, we continue to expect some normalization in DRAM supply/demand and thus price. We model market DRAM price to go through a steady drop throughout 2028. We don't expect CXMT to escape the market cyclicity, but we think the support from localization demand will help its price more resilient ([Exhibit 36](#)).
- Furthermore, as mentioned earlier, we expect CXMT to put more capacity into HBM. HBM is of national strategic importance and we believe HBM pricing will be more stable & allow CXMT to earn decent margins. HBM does not generate revenue for CXMT today, but will contribute more and more especially in 2028. This “catch-up” in HBM hence should dampen the impact of falling conventional DRAM price and make CXMT more resilient in 2028 too. Overall we model CXMT's revenue to surge 633% and 63% in 2026 and 2027, and drop slightly by 6% in 2028 ([Exhibit 37](#)).

EXHIBIT 36: We expect CXMT's DRAM ASP to trend largely together with global DRAM pricing, but possibly stay more resilient supported by localization demand.



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 37: We forecast CXMT's revenue to surge 633% and 63% in 2026 and 2027, before dropping slightly by 6% in 2028.



Source: Company reports, Bernstein estimates and analysis

CXMT's margin has been surprisingly good and largely on par with the incumbents' now. The ability to weather downturns is likely similar too.

- CXMT was significantly behind leaders in profitability before. It was mostly due to high fixed cost burden but as its scale increased the gap was narrowing quickly. Also thanks to surging pricing, CXMT's margins improved remarkably from 2H25 and as of 2Q26 was pretty much on par with the global incumbents' ([Exhibit 38-Exhibit 39](#)). This is despite being behind in process nodes and was a surprise to us. We think main reason is likely the lack of HBM in CXMT's sales mix now, which actually is a margin drag today compared to conventional DRAM. Apart from that we also suspect CXMT may enjoy some cost advantage in construction and operation. We also highlight that if we exclude D&A and look at EBITDA margin, CXMT was already approaching the incumbents in 2024 and roughly in line in 1H25 ([Exhibit 40](#)). All these suggest even before the DRAM price upcycle, CXMT likely was already competitive on cash cost. More importantly, similar EBITDA margin implies similar cash cost, and hence that CXMT's ability to weather future downturns is likely similar to the global incumbents too.

Aggressive capex will create a cost burden, but the drag should be minimum vs. the elevated memory prices.

- Going forward we model gross margin to improve marginally in 2H26 and stay elevated at around 90% throughout 2027. Margins should decline in 2028 along with ASP but should remain above 80% for GM and 70% for OPM by 4Q28 ([Exhibit 41](#)). We think cost reduction per year will decelerate from prior years but remain decent in 2026 and 2027, driven by G4 yield improvement. Notably we model average cost per bit to increase in 2028 due to heavier D&A burden and more importantly more wafers allocated to HBM production ([Exhibit 42](#)). That said price is still more important than cost & will remain the much bigger driver of margins then.

We don't model dividend or buyback, and assume CXMT to preserve ample cash to expand future capacity regardless of cycles.

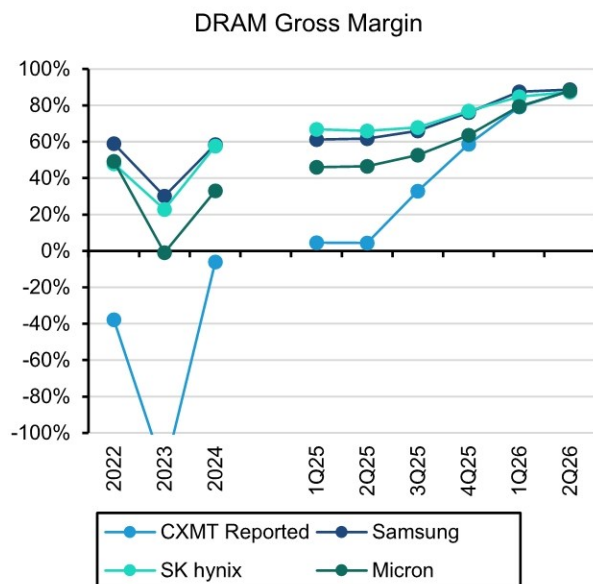
- CXMT's Articles of Incorporation does stipulate a minimum of 10% of the company's annual distributable profit must be paid back to shareholders as cash dividends. Nonetheless, the requirement is conditional on the firm not expecting "major capital expenditure". And we think the rapidly rising capex going forward likely can qualify as being major and spare CXMT from having to pay anything. So our model does not model any dividend payout up to 2028. Even if CXMT does pay anything we think it likely will be at the minimum 10% level and not very material for investors.

- On a high level CXMT's long term goal of achieving DRAM self-sufficiency for China would require significant and continued capex for many years. As a result we think CXMT would prefer to keep cash in hand so that it can execute its capacity plan without disruption from memory cycles. In 2H26 earnings online Q&A, CXMT also indicated that it would prioritize on capex as the company remains in capacity expansion phase.

Share of minority interest has seen drastic changes lately and presents significant uncertainty to our EPS forecast.

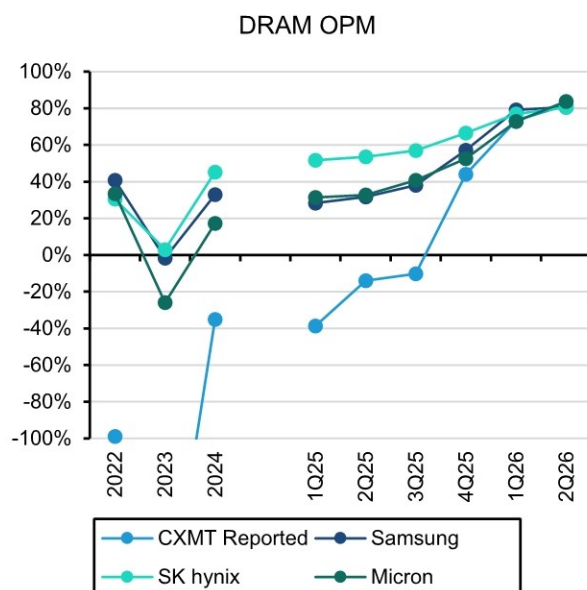
- CXMT took funding from government when building its fabs in earlier years. Specifically Hefei Fab 2 and Beijing Fab 1&2 both relied on funding from external parties. CXMT actually has only around 30% stake in these fabs, but exerts control over them through acting-in-concert agreement with other major shareholders, and hence is able to consolidate these fabs into its financial statements. That is why minority interests' share of net profit/loss increased meaningfully from 2022 to 2025 as these non-wholly owned fabs ramped up and took a bigger share of the overall profit. Minority interests' share in consolidated profit reached 63% in 2H25, reflecting the fact that CXMT only has ~30% stake in these fabs which take up 2/3 of CXMT's wafer capacity and possibly a bigger share of bit output ([Exhibit 43](#)).
- However, minorities' share fell drastically to 25-29% in 1H26 and we were not sure why. It likely involved some inter-subsidiary transactions but we couldn't find any specifics from the disclosures. This makes our forecast uncertain. Furthermore, capital structure of future fabs, whether they will be wholly owned by CXMT parent, or continue to involve minority shareholders, is also unknown and will have an impact on minorities' share of profit and therefore EPS forecast of CXMT too. In our forecast we choose to assume that minority's share stay at 25% going forward & highlight that as a major risk to our EPS projection.

EXHIBIT 38: CXMT's margin was behind by a wide margin before 2025...



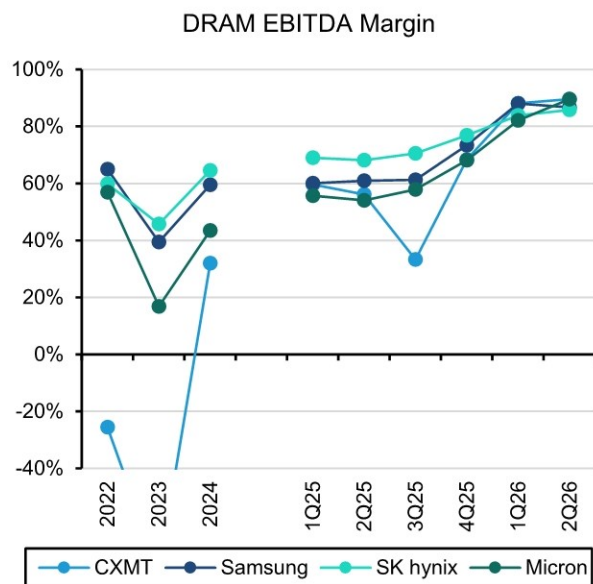
Source: Company reports and Bernstein analysis

EXHIBIT 39: ... but has been improving quickly and is now roughly on par with incumbents'



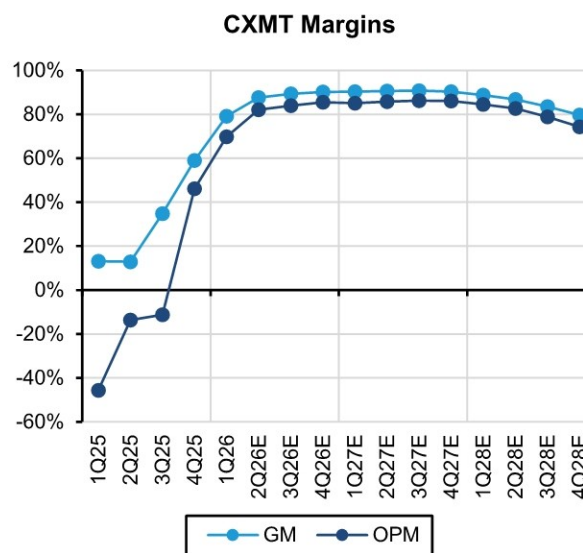
Source: Company reports and Bernstein analysis

EXHIBIT 40: CXMT's cash margin was already similar to the incumbents' in 1H25, before the DRAM price surge.



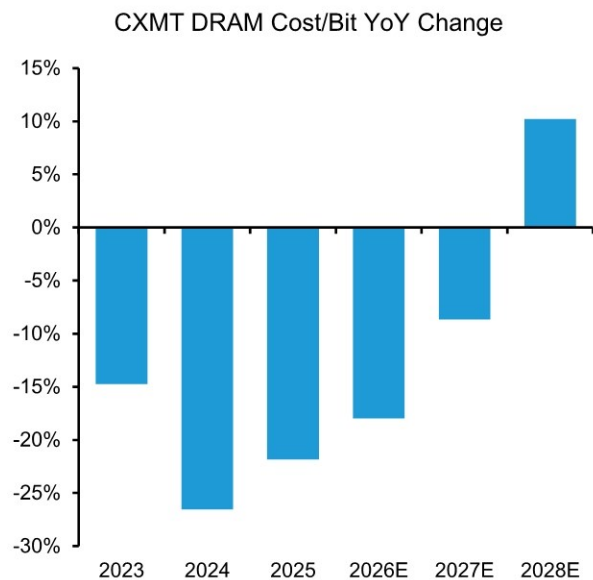
Source: Company reports and Bernstein analysis

EXHIBIT 41: We forecast CXMT's margins to further increase in 2H26 and remain elevated in 2027, before moderating in 2028.



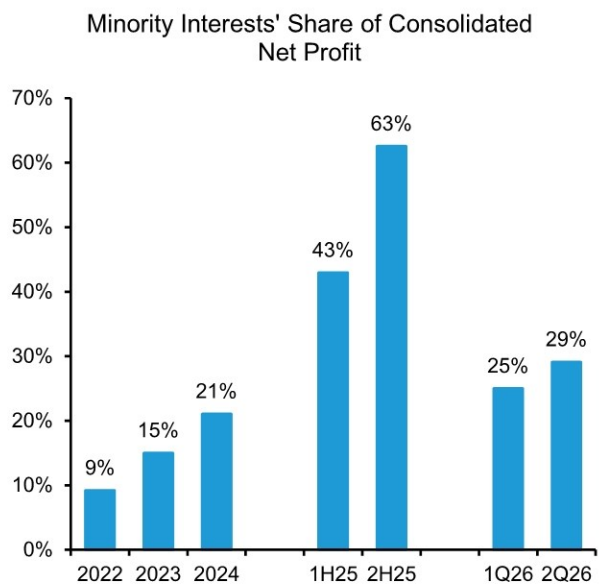
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 42: We expect bit cost reduction to continue but reverse in 2028 due to higher HBM mix.



Source: Company reports, Bernstein estimates and analysis

EXHIBIT 43: CXMT's share of profit attributable to minority changed dramatically in 1H26 and is a source of uncertainty for future EPS forecast.



Source: Company reports and Bernstein analysis

Our forecast is notably above consensus in the next 2 years but roughly in line in 2028 as we expect DRAM prices to fall.

- We forecast CXMT's revenue to rise 65% YoY in 2027 as the industry continues to enjoy the elevated cycle, and CXMT outgrows the industry. The current cycle is expected to peak toward the end of 2027, and "normalize" in 2028, but localization policy & catching-up in HBM will make the normalization milder for CXMT. We model CXMT's revenue to dip 6%

in 2028 ([Exhibit 37](#)). Margins will follow a similar trajectory ([Exhibit 41](#)). EPS is modeled to grow 70% YoY in 2027 & dip 12% in 2028, as operating leverage/deleverage results in a slightly bigger swing from revenue to earnings ([Exhibit 44-Exhibit 45](#)).

- Versus consensus, we are 20-30% above in revenue and 40-50% above in EPS for 2026 and 2027. We think consensus has not fully reflected 1H26's strong results and may be underestimating CXMT's bit growth as well. Our 2028 forecast is slightly behind consensus in revenue but in line in EPS, mainly because we have a relatively more conservative view on cycle and sustainability of DRAM price ([Exhibit 46](#)).

EXHIBIT 44: We forecast CXMT's EPS to rise dramatically this year and rise further in 2027, before moderating in 2028.

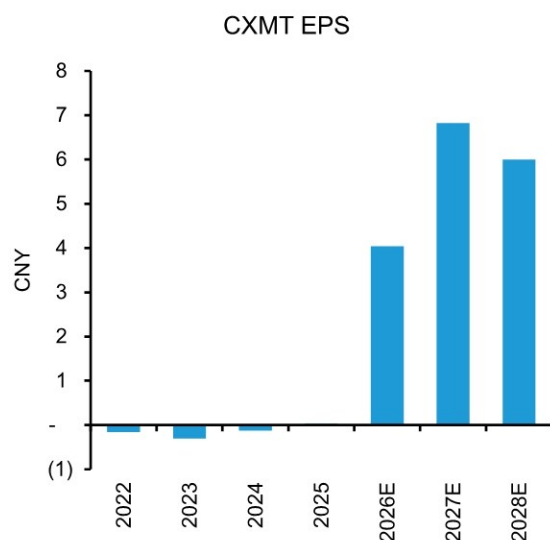
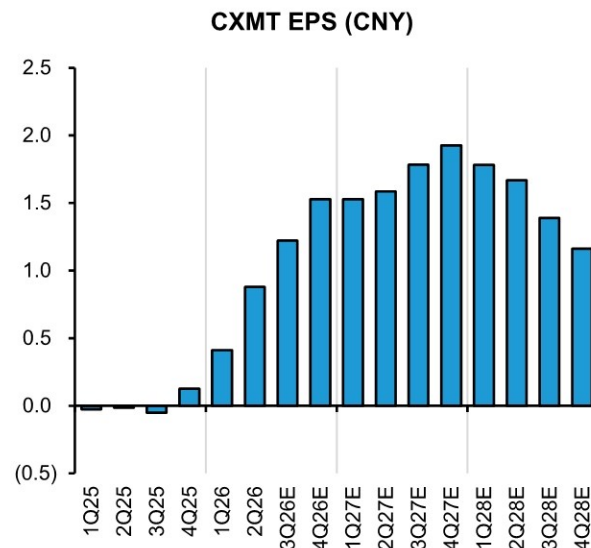


EXHIBIT 45: We project EPS to keep rising and peak towards the end of 2027.



Source: Company reports, Bernstein estimates and analysis

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 46: We are notably above consensus for 2026 and 2027 but more conservative in 2028 due to our more cautious view on DRAM cycle and pricing.

CXMT	2026E			2027E			2028E		
	Bernstein New	Consensus	% Diff	Bernstein New	Consensus	% Diff	Bernstein New	Consensus	% Diff
Revenue (CNY B)	452.9	361.4	25.3%	749.4	587.4	27.6%	702.4	762.2	-7.8%
YoY	633%	485%		65.5%	62.6%		-6.3%	29.7%	
Gross Margin	88.1%	85.4%	2.7%	90.5%	85.5%	5.0%	85.1%	84.4%	0.7%
Operating Margin	82.5%	73.9%	8.6%	85.8%	75.8%	10.0%	80.5%	74.2%	6.3%
EPS (CNY)	4.0	2.8	43.9%	6.8	4.6	48.5%	6.0	5.8	3.8%
YoY	12328%	8534%		68.9%	63.6%		-12.1%	25.9%	

Source: Bloomberg, Bernstein estimates and analysis

VALUATION, RATING & RISK

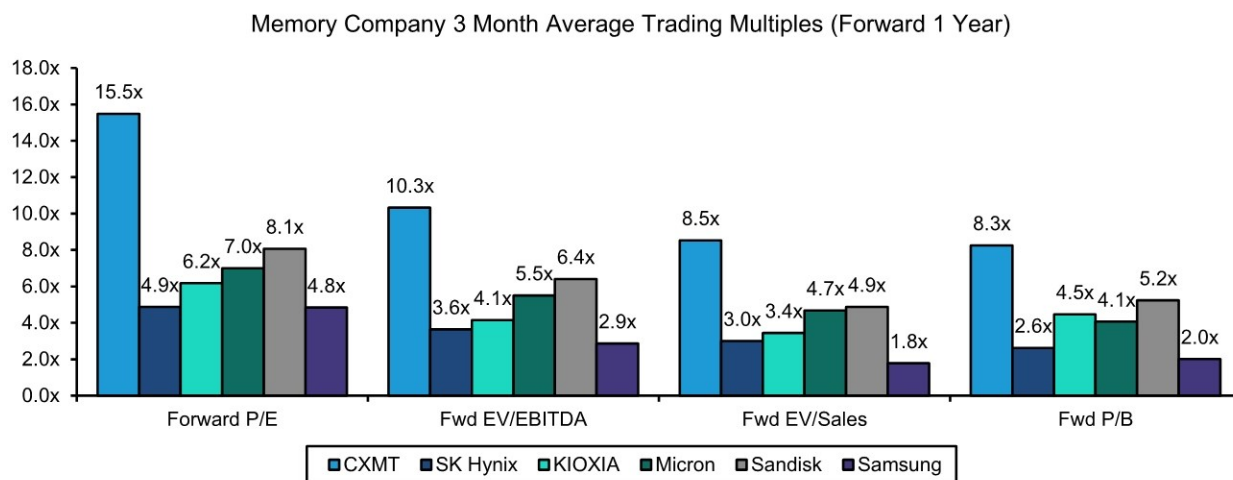
Compared to global peers, CXMT trades at a higher multiple, but arguably justified by China's localization demand.

- CXMT stock went up ~6x on the IPO day and then has mostly been range bound between CNY50-60 in the past ~8 weeks since its IPO. It currently trades at 13x 1-year forward P/E and 7.3x 1-year forward P/B. And it does appear notably more expensive than overseas peer memory makers on all the common valuation metrics ([Exhibit 47-Exhibit 48](#)).
- The lack of EUV clouding its long-term technology runway should make CXMT's valuation lower, but DUV is sufficient to support CXMT to grow for a few years. CXMT is subject to cyclicity but China's localization policy, especially on HBM, should make a downcycle more benign for CXMT. More importantly, A-share investors have a different appetite and a difficult valuation framework and semiconductor stocks usually enjoy higher valuation that their overseas counterparts can't have.
- We think this is the reason that has been supporting CXMT's valuation premium. While it is clearly more expensive than memory stocks in other countries, we expect this higher valuation to persist.

More importantly, CXMT is actually cheaper than other A-share semiconductor stocks.

- We also compare CXMT with A-share semiconductor peers in valuation. We select prominent A-share listed companies into four groups - foundry, memory related, semi equipment & AI GPU/CPU ([Exhibit 49](#)) - and find their average valuations. As shown in [Exhibit 50](#), CXMT is considerably cheaper in most cases.
- One can argue it is because memory is in a peak cycle and has peak earnings currently. We won't disagree, but even when we look at forward 2-year earnings, CXMT remains notably cheaper ([Exhibit 51](#)). Even if CXMT's 2-year forward EPS halves vs. consensus, its P/E ratio would still appear very low vs. nearly all the domestic peers.

EXHIBIT 47: CXMT trades at a notable premium over overseas memory companies on all common metrics.

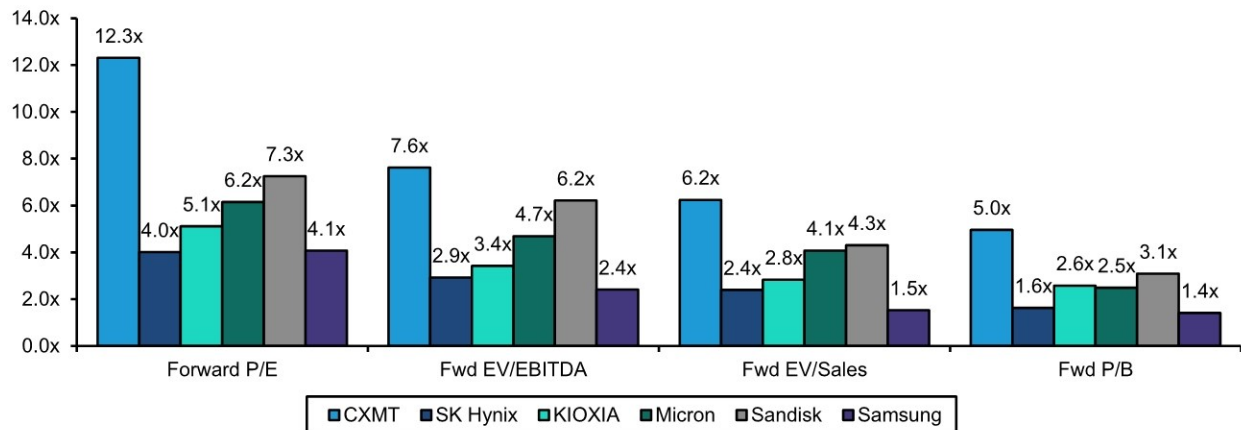


3 months up to Sep 11

Source: Bloomberg and Bernstein analysis

EXHIBIT 48: **The valuation premium over overseas peers exists in 2-year forward multiples too.**

Memory Company 3 Month Average Trading Multiples (Forward 2 Year)



3 months up to Sep 11

Source: Bloomberg and Bernstein analysis

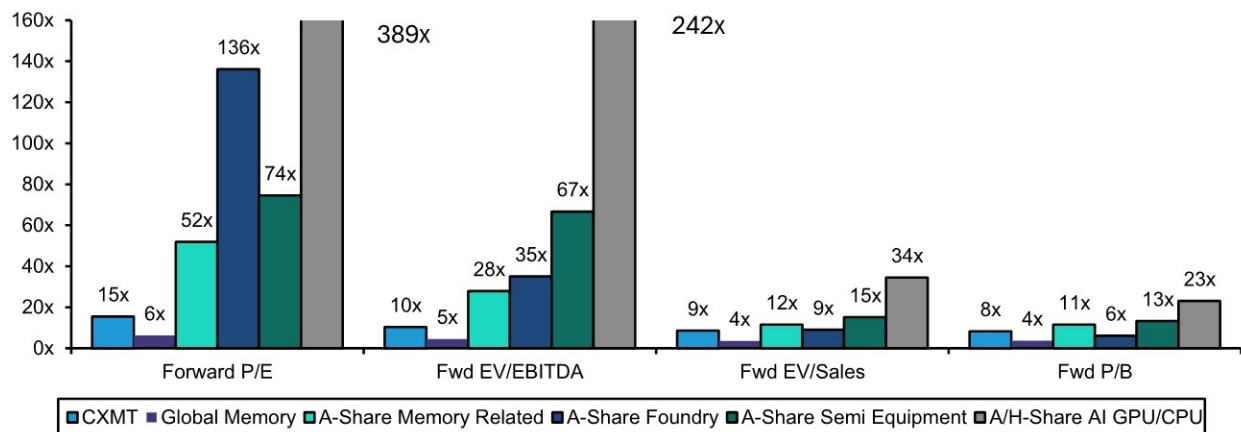
EXHIBIT 49: **We select prominent A-share listed semiconductor companies into four groups to find their average valuations.**

Foundry			Memory Related			Semi Equipment			AI GPU/CPU		
SMIC	中芯国际	688981 CH	GigaDevice	兆易创新	603986 CH	AMEC	中微公司	688012 CH	Hygon	海光信息	688041 CH
Hua Hong	华虹公司	688347 CH	Montage	澜起科技	688008 CH	NAURA	北方华创	002371 CH	Iluvatar	天数智芯	9903 HK
Nexchip	晶合集成	688249 CH	Ingenic	北京君正	300223 CH	Piotech	拓荆科技	688072 CH	MetaX	沐曦股份	688802 CH
CR Micro	华润微	688396 CH	Puran Semi	普冉股份	688766 CH	ACMR	盛美上海	688082 CH	Biren	壁仞科技	6082 HK
			Giantec	聚辰股份	688123 CH				Cambricon	寒武纪	688256 CH
			Longsys	江波龙	301308 CH						
			Biwin	佰维存储	688525 CH						
			Demingli	德明利	001309 CH						
			Shannon Semi	香农芯创	300475 CH						
			Netac	朗科科技	300042 CH						
			Shenzhen Kaifa	深科技	000021 CH						

Source: Bernstein analysis

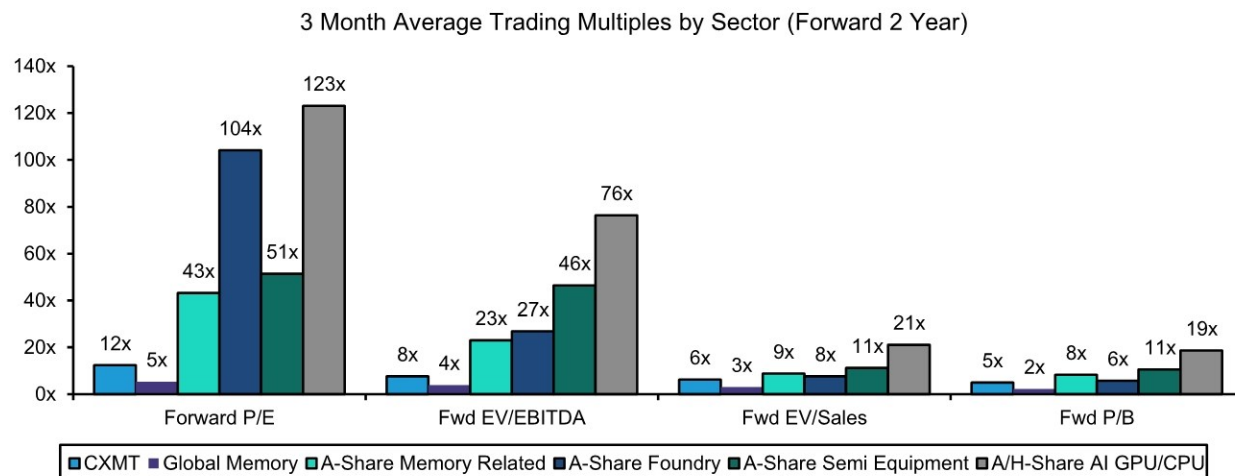
EXHIBIT 50: **CXMT is actually cheaper than other A-share listed semiconductor stocks.**

3 Month Average Trading Multiples by Sector (Forward 1 Year)



3 months up to Sep 11

Source: Bloomberg and Bernstein analysis

EXHIBIT 51: **On 2 year forward basis, CXMT remains much cheaper as well.**

3 months up to Sep 11

Source: Bloomberg and Bernstein analysis

We set our 1-year price target at CNY 70, on 10x of Q5-Q8 EPS, equivalent to 4.4x 2-year forward P/B, & rate CXMT Outperform.

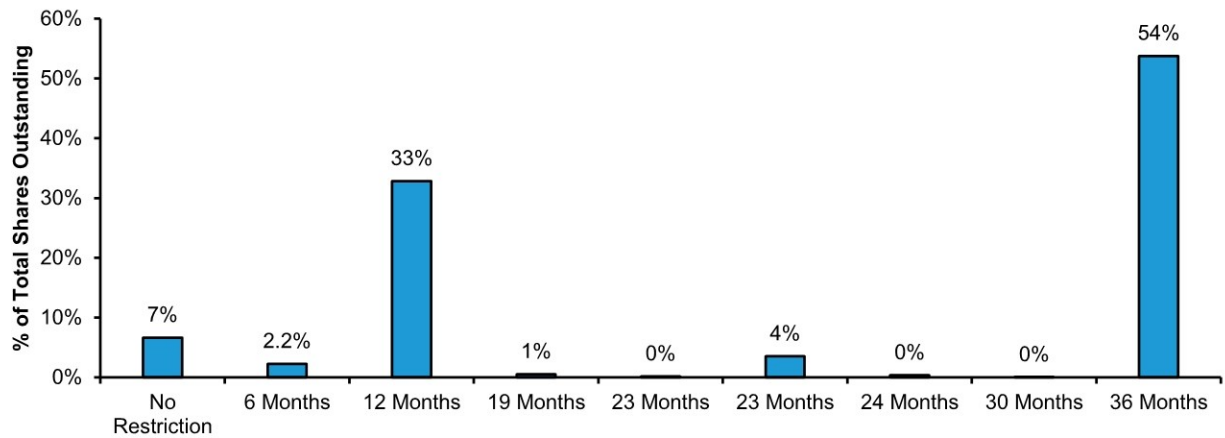
- Using P/E to value other memory stocks too, we value CXMT at 10x blended 2-year forward EPS (forward 5Q-8Q) and arrive at our price target at CNY 70. This represents 31% upside as of Sep 17. The target P/E is moderately below the current P/E ratio as we consider EPS is at a peak cycle. Our price target corresponds to 4.4x 2-year forward P/B and 6.2x forward EV/EBITDA, which is above global memory incumbents', but again considerably below A-share semiconductor peers'.

Investors should beware of the possible impact when a significant number of shares come out of lock-up period in Jan & much more meaningfully in July 2027.

- After IPO, most of CXMT's outstanding shares are still in a lock-up. Currently only 6.7% of shares is in free float and most of that is shares issued to retail investors through online offering. In 6 months post IPO (Jan 27, 2027), 2.2% of the total shares will come out of lock-up, which is a small part of the total outstanding shares but a significant increase (34%) from the current free float ([Exhibit 52](#)). And all of that are from institutional investors that acquired the shares in IPO. We think this probably will lead to some market impact and caution investors to take note. A significantly larger part (33% of the total outstanding shares) will come out of the restriction 12 months after the IPO (Jul 27, 2027), and they mostly consist of strategic investors (GigaDevice, Alibaba, Tencent, etc.) and private equity investors. Hence, they likely will not dispose of all of their stakes immediately but the sheer size of it still warrants attention from investors.

EXHIBIT 52: There will be a sizable amount of shares coming out of lock-up period in 6 and 12 months after IPO and investors should beware of the potential market impact.

Distribution of CXMT Stock Lock-Up Period Post IPO



Source: Company reports, WIND and Bernstein analysis

APPENDIX - FINANCIAL FORECASTS

EXHIBIT 53: CXMT Income Statement

Figures in CNY Mn Unless Otherwise Stated

Income Statement	2022	2023	2024	2025	2026E	2027E	2028E	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E
Revenue	8,287	9,087	24,178	61,799	452,851	749,441	702,371	50,800	99,510	132,173	170,368	169,190	174,252	194,946	211,054
COGS	8,546	9,263	22,830	36,465	53,941	71,079	104,613	10,588	12,346	14,173	16,833	16,360	16,366	17,908	20,444
Gross Profit	(259)	(175)	1,349	25,334	398,910	678,363	597,758	40,212	87,163	118,000	153,534	152,830	157,885	177,038	190,609
SG&A	977	2,446	2,582	6,968	5,052	8,585	8,046	561	1,025	1,514	1,952	1,938	1,996	2,233	2,418
R&D	2,486	4,520	4,607	9,593	16,788	25,182	27,701	2,154	3,383	5,000	6,251	6,000	6,000	6,500	6,682
Net Financial Expense	1,796	1,934	2,204	2,954	4,257	(1,005)	(6,704)	1,192	1,238	1,151	677	305	(59)	(429)	(823)
Other Operating Exp. (Inc.)	3,681	10,265	924	(1,902)	(906)	2,463	3,148	872	(194)	(678)	(906)	569	581	834	679
Operating Income	(9,210)	(19,341)	(8,969)	7,721	373,719	643,137	565,567	35,433	81,711	111,013	145,562	144,017	149,368	168,098	181,652
Other Inc/(Exp)	40	118	(80)	(20)	0	(2)	(2)	1	(0)	(0)	(0)	(0)	(0)	(0)	(1)
Profit Before Tax	(9,170)	(19,223)	(9,049)	7,701	373,719	643,135	565,565	35,434	81,711	111,013	145,561	144,017	149,368	168,098	181,652
Income Tax/(Benefit)	1	2	2	557	23,535	25,725	22,623	2,423	7,173	6,661	7,278	5,761	5,975	6,724	7,266
Net Income	(9,171)	(19,225)	(9,051)	7,144	350,185	617,410	542,942	33,012	74,538	104,352	138,283	138,256	143,393	161,374	174,386
Minority Interest	(843)	(2,885)	(1,906)	5,269	90,603	154,352	135,736	8,249	21,695	26,088	34,571	34,564	35,848	40,344	43,597
Net Profit Attributable to Parent	(8,328)	(16,340)	(7,145)	1,875	259,581	463,057	407,207	24,762	52,843	78,264	103,712	103,692	107,545	121,031	130,790
Basic EPS (CNY)	(0.16)	(0.30)	(0.13)	0.03	4.04	6.82	6.00	0.41	0.88	1.22	1.53	1.53	1.58	1.78	1.93
Diluted EPS (CNY)	(0.16)	(0.30)	(0.13)	0.03	4.04	6.82	6.00	0.41	0.88	1.22	1.53	1.53	1.58	1.78	1.93
Basic WASO	52,843	53,626	56,571	57,688	63,077	67,884	67,884	60,193	60,193	64,038	67,884	67,884	67,884	67,884	67,884
Diluted WASO	52,843	53,626	56,571	57,688	63,077	67,884	67,884	60,193	60,193	64,038	67,884	67,884	67,884	67,884	67,884
IFRS Adjusted Operating Income	(8,188)	(19,030)	(8,496)	7,668	372,396	636,941	554,663	37,229	81,435	110,131	143,600	143,162	148,106	166,316	179,357
EBITDA	(2,120)	(6,910)	7,728	34,755	404,249	681,268	622,585	44,759	89,148	118,048	152,294	152,884	158,688	177,784	191,911
Margins															
Gross Margin	(3.1%)	(1.9%)	5.6%	41.0%	88.1%	90.5%	85.1%	79.2%	87.6%	89.3%	90.1%	90.3%	90.6%	90.8%	90.3%
EBITDA Margin	(25.6%)	(76.3%)	32.0%	56.2%	89.3%	90.9%	88.6%	88.1%	89.6%	89.3%	88.4%	90.4%	91.1%	91.2%	90.9%
Operating Margin	(111.1%)	(212.8%)	(37.1%)	12.5%	82.5%	85.8%	80.5%	69.7%	82.1%	84.0%	85.4%	85.1%	85.7%	86.2%	86.1%
IFRS Adjusted OPM	(98.8%)	(209.4%)	(35.1%)	12.4%	82.2%	85.0%	79.0%	73.3%	81.8%	83.3%	84.3%	84.6%	85.0%	85.3%	85.0%
Net Income Margin	(110.7%)	(211.6%)	(37.4%)	11.6%	77.3%	82.4%	77.3%	65.0%	74.9%	79.0%	81.2%	81.7%	82.3%	82.8%	82.6%
ROE		(38.3%)	(18.6%)	3.8%	118.0%	75.3%	38.8%	143.1%	195.3%	151.1%	125.2%	95.3%	79.5%	73.9%	67.0%
Growth															
Revenue Growth		9.7%	166.1%	155.6%	633%	65.5%	(6.3%)	71.0%	95.9%	32.8%	28.9%	(0.7%)	3.0%	11.9%	8.3%
Operating Profit Growth		N.A.	N.A.	N.A.	4740%	72.1%	(12.1%)	158.8%	130.6%	35.9%	31.1%	(1.1%)	3.7%	12.5%	8.1%
EPS Growth		N.A.	N.A.	N.A.	12328%	68.9%	(12.1%)	228.2%	113.4%	39.2%	25.0%	(0.0%)	3.7%	12.5%	8.1%
Key Operating Metrics	2022	2023	2024	2025	2026E	2027E	2028E	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E
Total Revenue	8,287	9,087	24,178	61,799	452,851	749,441	702,371	50,800	99,510	132,173	170,368	169,190	174,252	194,946	211,054
DRAM Revenue (CNY M)	7,891	8,583	22,972	60,235	447,178	741,947	695,347	49,889	97,774	130,851	168,664	167,498	172,509	192,997	208,943
DRAM Revenue (US\$M)	1,171	1,211	3,190	8,431	66,013	110,081	103,167	7,204	14,371	19,414	25,024	24,851	25,595	28,635	31,000
Gb Shipped (M Gb) (inc HBM)	3,081	5,946	10,220	20,104	36,212	51,802	69,981	6,466	8,082	9,727	11,938	11,700	11,933	13,372	14,796
ASP (\$/Gb) (inc HBM)	0.38	0.20	0.31	0.42	1.82	2.13	1.47	1.11	1.78	2.00	2.10	2.12	2.14	2.14	2.10
Cost (\$/Gb) (inc HBM)	0.52	0.45	0.33	0.26	0.21	0.19	0.21	0.23	0.21	0.21	0.20	0.20	0.19	0.19	0.19
Gb Shipped Change		93.0%	71.9%	96.7%	80.1%	43.0%	35.1%	(5.0%)	25.0%	20.3%	22.7%	(2.0%)	2.0%	12.1%	10.7%
ASP Change		(46.4%)	53.3%	34.3%	334.7%	(16.6%)	(30.6%)	84.7%	59.6%	12.3%	5.0%	1.3%	1.0%	(0.2%)	(2.2%)
Cost Change		(13.9%)	(26.7%)	(22.2%)	(19.3%)	(8.7%)	(10.2%)	(6.9%)	(7.6%)	(1.5%)	(0.7%)	(1.0%)	(2.1%)	(2.5%)	(2.6%)
DRAM Gross Margin	(37.8%)	(121.4%)	(6.0%)	38.6%	88.4%	90.9%	85.6%	79.2%	88.0%	89.4%	90.5%	90.7%	91.0%	91.2%	90.7%

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 54: CXMT Balance Sheet and Cash Flow Statement

Figures in CNY Mn Unless Otherwise Stated

Balance Sheet															
	2022	2023	2024	2025	2026E	2027E	2028E	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E
Cash & Cash Equivalents	32,072	31,374	42,699	51,990	396,204	886,445	1,237,664	84,735	143,426	289,072	396,204	505,809	621,978	753,087	886,445
Accounts & Notes Receivable	124	399	1,846	1,524	9,259	11,470	8,012	2,915	6,899	7,183	9,259	9,399	9,574	10,595	11,470
Inventories	6,917	10,736	18,550	28,568	45,500	53,072	82,546	31,883	35,130	39,090	45,500	44,298	42,952	46,488	53,072
Other Current Assets	3,156	4,941	4,920	28,904	51,533	51,731	46,392	39,319	47,161	49,256	51,533	50,863	50,626	51,361	51,731
Total Current Assets	42,268	47,450	68,014	110,986	502,495	1,002,718	1,374,614	158,853	232,616	384,601	502,495	610,370	725,131	861,531	1,002,718
PP&E	94,664	136,256	194,461	214,937	263,724	362,451	531,916	217,057	219,847	238,770	263,724	283,145	302,563	326,562	362,451
Other Non-Current Assets	10,761	9,050	9,123	10,862	15,466	15,656	15,628	12,257	15,615	15,477	15,466	15,457	15,468	15,541	15,656
Non-Current Assets	105,426	145,307	203,585	225,799	279,191	378,107	547,544	229,314	235,462	254,246	279,191	298,602	318,031	342,102	378,107
Total Assets	147,694	192,756	271,599	336,785	781,686	1,380,825	1,922,158	388,166	468,078	638,848	781,686	908,972	1,043,162	1,203,634	1,380,825
Debt	65,815	91,822	97,795	128,513	99,979	69,432	49,390	130,141	120,815	109,849	99,979	91,097	83,102	75,907	69,432
Accounts Payable	3,296	4,443	6,416	9,149	16,581	20,138	29,524	13,426	12,295	13,960	16,581	16,473	16,298	17,640	20,138
Other Liabilities	14,366	31,377	63,131	45,023	85,135	93,855	102,902	52,994	64,218	73,330	85,135	83,155	82,121	87,072	93,855
Total Liabilities	83,478	127,643	167,342	182,685	201,695	183,424	181,815	196,561	197,329	197,140	201,695	190,725	181,522	180,619	183,424
Shareholders' Equity	49,672	35,995	41,420	56,754	383,305	846,362	1,253,569	81,724	134,722	279,593	383,305	486,997	594,542	715,573	846,362
Minority Interest	14,545	29,518	62,837	97,346	196,686	351,039	486,774	109,881	136,027	162,115	196,686	231,250	267,098	307,442	351,039
Total Equity	64,216	65,113	104,257	154,100	579,991	1,197,401	1,740,343	191,605	270,749	441,708	579,991	718,247	861,640	1,023,015	1,197,401
Total Liabilities & Equity	147,694	192,756	271,599	336,785	781,686	1,380,825	1,922,158	388,166	468,078	638,848	781,686	908,972	1,043,162	1,203,634	1,380,825
BVPS (CNY)	1.03	0.66	0.72	0.94	5.65	12.47	18.47	1.36	2.24	4.12	5.65	7.17	8.76	10.54	12.47
Cash Flow Statement															
	2022	2023	2024	2025	2026E	2027E	2028E	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E
Net Income	(9,171)	(19,225)	(9,051)	7,144	350,185	617,410	542,942	33,012	74,538	104,352	138,283	138,256	143,393	161,374	174,386
Depreciation & Amortization	6,068	12,120	16,224	27,086	31,853	44,327	67,921	7,529	7,713	7,917	8,695	9,722	10,582	11,468	12,554
Other	1,146	(167)	(276)	2,289	16,683	2,726	(1,703)	2,025	6,339	4,547	3,772	(246)	307	1,107	1,558
CF From Operations	(1,957)	(7,272)	6,897	36,520	398,720	664,463	609,161	42,566	88,590	116,816	150,749	147,732	154,282	173,590	188,499
Capex	(35,431)	(43,658)	(71,230)	(49,739)	(82,153)	(143,674)	(237,900)	(9,394)	(12,201)	(26,811)	(33,747)	(29,244)	(30,119)	(35,645)	(48,666)
Other	8,482	(343)	(1,186)	(37,084)	(13,983)	-	-	(6,389)	(7,594)	-	-	-	-	-	-
CF From Investing	(26,949)	(44,002)	(72,415)	(86,823)	(96,135)	(143,674)	(237,900)	(15,783)	(19,794)	(26,811)	(33,747)	(29,244)	(30,119)	(35,645)	(48,666)
Equity Issued	3,811	137	10,800	6,856	66,614	-	-	6	1	66,607	-	-	-	-	-
Investment From Minority Shareholder	8,221	17,903	35,067	24,939	8,556	-	-	4,185	4,371	-	-	-	-	-	-
Debt Issued (repaid)	38,078	31,876	27,760	15,277	(25,201)	(30,548)	(20,042)	5,192	(9,557)	(10,966)	(9,870)	(8,883)	(7,994)	(7,195)	(6,475)
Other	(11,139)	891	3,587	(951)	(3,195)	-	-	(1,119)	(2,078)	-	-	-	-	-	-
CF From Financing	38,971	50,807	77,214	46,121	46,773	(30,548)	(20,042)	8,263	(7,261)	55,641	(9,870)	(8,883)	(7,994)	(7,195)	(6,475)
Cash at Beginning of Period															
Net Cash Flow	21,606	32,072	31,374	42,699	51,990	396,204	886,445	55,907	84,735	143,426	289,072	396,204	505,809	621,978	753,087
Adjustments	10,665	(466)	11,696	(4,182)	349,358	490,241	351,219	31,044	64,645	145,645	107,132	109,605	116,169	131,108	133,358
Free Cash Flow	402	(233)	13,474	(8,364)	(5,843)	0	0	(8,601)	(2,534)	0	0	0	0	0	0
Cash at End of Period															
Free Cash Flow	32,072	31,374	42,699	51,990	396,204	886,445	1,237,664	84,735	143,426	289,072	396,204	505,809	621,978	753,087	886,445
Free Cash Flow	(37,388)	(50,930)	(64,332)	(13,219)	316,568	520,789	371,261	33,172	76,389	90,005	117,002	118,488	124,163	138,304	139,833

BERNSTEIN TICKER TABLE

			18 Sep 2026		TTM	Reported EPS				Reported P/E (x)		
			Closing	Price	Rel.							
Ticker	Rating	Cur	Price	Target	Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
688825.CH (CXMT)	O	CNY	55.54	70.00	NA	CNY	0.03	4.04	6.82	N/M	13.8	8.1
ASIAx			1,953.49									

COVERAGE INITIATION

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY**CXMT Corp**

We value CXMT at 10x blended 2-year forward EPS (forward 5Q-8Q) of CNY7.16 and arrive our price target of CNY 70.

RISKS**CXMT Corp**

The downside risk to our target price includes 1) increased challenges in acquiring semiconductor equipment needed for capacity expansion, 2) faster-than-expected expansion in global DRAM supply 3) significant slow down in AI-related DRAM demand causing shortage to revert into oversupply and DRAM prices to decrease

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- **Credit Outperform (C-OP):** The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

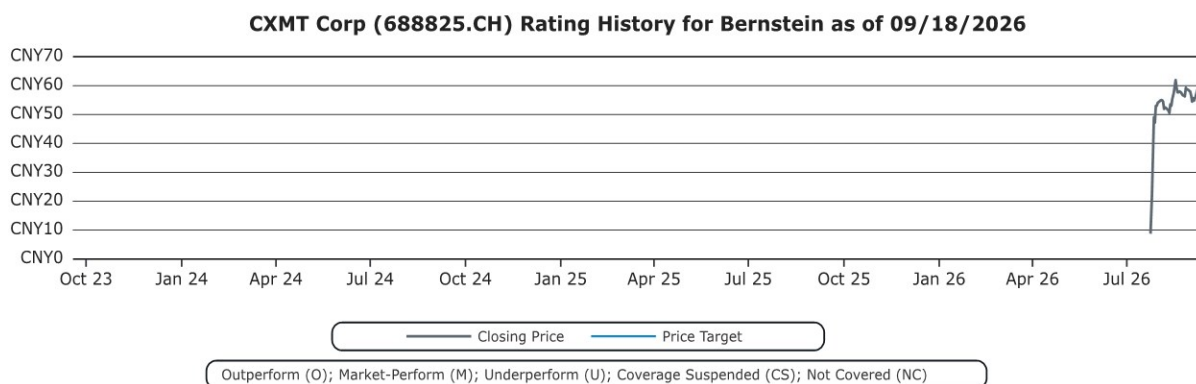
DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY



All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

OTHER MATTERS

The legal entity(ies) employing the analyst(s) listed in this report, and their location, can be determined by the country code of their phone number, as follows:

- +1 Bernstein Institutional Services LLC; New York, New York, USA
- +44 Bernstein Autonomous LLP; London UK
- +212 Société Générale Global Solution Center Morocco, Casablanca, Morocco
- +33 BSG France S.A.; Paris, France
- +34 BSG France S.A.; Madrid, Spain
- +41 Bernstein Autonomous LLP; Geneva, Switzerland
- +49 BSG France S.A.; Frankfurt, Germany
- +91 Sanford C. Bernstein (India) Private Limited; Mumbai, India
- +852 Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司; Hong Kong, China
- +65 Sanford C. Bernstein (Singapore) Private Limited; Singapore
- +81 Sanford C. Bernstein Japan KK; Tokyo, Japan

Where this report has been prepared by research analyst(s) employed by a non-US affiliate, such analyst(s), is/are (unless otherwise expressly noted below) not registered as associated persons of Bernstein Institutional Services LLC or any other SEC-registered broker-dealer and are not licensed or qualified as research analysts with FINRA. Accordingly, such analyst(s) may not be subject to FINRA's restrictions regarding (among other things) communications by research analysts with a subject company, interactions between research analysts and investment banking personnel, participation by research analysts in solicitation and marketing activities relating to investment banking transactions, public appearances by research analysts, and trading securities held by a research analyst account.

Where this report has been prepared by research analyst(s) employed by Société Générale Africa Technologies & Services (part of the Société Générale group of companies), it has been prepared on behalf of a Bernstein company under a Global Services Agreement in place between Bernstein and Société Générale.

CERTIFICATION

Each research analyst listed in this report, who is primarily responsible for the preparation of the content of this report, certifies that all of the views expressed in this publication accurately reflect that analyst's personal views about any and all of the subject securities or issuers and that no part of that analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this publication.

II. ADDITIONAL GLOBAL CONFLICT DISCLOSURES

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e., the private side) within the Firm, and into other areas, units, groups or affiliates (i.e., public side) of the Firm.

III. OTHER IMPORTANT INFORMATION AND DISCLOSURES

Separate branding is maintained for "Bernstein" and "Autonomous" research products.

- Bernstein produces a number of different types of research products including, among others, fundamental analysis and quantitative analysis under both the "Autonomous" and "Bernstein" brands. Recommendations contained within one type of research product may differ from recommendations contained within other types of research products, whether as a result of differing time horizons, methodologies or otherwise. Furthermore, views or recommendations within a research product issued under one brand may differ from views or recommendations under the same type of research product issued under the other brand. The Research Ratings System for the two brands and other information related to those Rating Systems are included in the previous section.
- Autonomous operates as a separate business unit within the following entities: Bernstein Institutional Services LLC, Bernstein Autonomous LLP, Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 and Sanford C. Bernstein (India) Private Limited. For information relating to "Autonomous" branded products (including certain Sales materials) please visit:

www.autonomous.com. For information relating to Bernstein branded products please visit: www.bernsteinresearch.com.

Analysts are compensated based on aggregate contributions to the research franchise as measured by account penetration, productivity and proactivity of investment ideas. No analysts are compensated based on performance in, or contributions to, generating investment banking revenues.

This report has been produced by an independent analyst as defined in Article 3 (1)(34)(i) of EU 596/2014 Market Abuse Regulation ("MAR") and the same article of MAR as it forms part of United Kingdom domestic law by virtue of the European Union (Withdrawal) Act 2018.

To our readers in the United States: Bernstein Institutional Services LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission ("SEC") and a member of the U.S. Financial Industry Regulatory Authority, Inc. ("FINRA") is distributing this publication in the United States and accepts responsibility for its contents. Where this material contains an analysis of debt product(s), such material is intended only for institutional investors and is not subject to the US independence and disclosure standards applicable to debt research prepared for retail investors.

Bernstein Institutional Services LLC may act as principal for its own account or as agent for another person (including an affiliate) in sales or purchases of any security which is a subject of this report. This report does not purport to meet the objectives or needs of any specific individuals, entities or accounts.

To our readers in Canada: If this publication pertains to a Canadian domiciled company, it is being distributed in Canada by Sanford C. Bernstein (Canada) Limited, which is licensed and regulated by the Canadian Investment Regulatory Organization. If the publication pertains to a non-Canadian domiciled company, it is being distributed by Bernstein Institutional Services LLC, which is licensed and regulated by both the SEC and FINRA, into Canada under the International Dealers Exemption.

This document may not be passed onto any person in Canada unless that person qualifies as "permitted client" as defined in Section 1.1 of NI 31-103.

To our readers in Brazil: This report has been prepared by Bernstein Institutional Services LLC, and Banco BTG Pactual S.A. ("BTG") is responsible for the distribution of this report in Brazil.

To readers in the United Kingdom: This publication has been issued or approved for issue in the United Kingdom by Bernstein Autonomous LLP, authorised and regulated by the Financial Conduct Authority and located at 60 London Wall, London EC2M 5SH, +44 (0)20-7170-5000. Registered in England & Wales No OC343985.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Financial Promotion Order"), (ii) are persons falling within Article 49(2)(a) to (d) ("high net worth companies, unincorporated associations, etc.") of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the FSMA) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as "relevant persons"). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

To our readers in the member states of the EEA: This publication is being distributed by BSG France SA, which is authorised and regulated by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and Autorité des Marchés Financiers (AMF).

To our readers in Hong Kong: This publication is being distributed in Hong Kong by Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, which is licensed and regulated by the Hong Kong Securities and Futures Commission (Central Entity No. AXC846) to carry out Type 4 (Advising on Securities) regulated activities and subject to the licensing conditions mentioned in the SFC Public Register (<https://www.sfc.hk/publicregWeb/corp/AXC846/details>). This publication is solely for professional investors, as defined in the Securities and Futures Ordinance (Cap. 571). The purpose of this report is solely to provide an analysis of the issuers referred to in this report and is not intended for any purpose contrary to the laws of Hong Kong.

To our readers in Singapore: This publication is being distributed in Singapore by Sanford C. Bernstein (Singapore) Private Limited, only to accredited investors or institutional investors, as defined in the Securities and Futures Act 2001 of Singapore ("SFA"). Recipients in Singapore should contact Sanford C. Bernstein (Singapore) Private Limited in respect of matters arising from, or in connection with, this publication. Sanford C. Bernstein (Singapore) Private Limited is regulated by the Monetary Authority of Singapore and licensed under the SFA as a capital markets services licence holder for dealing in capital markets products that are securities and collective investment schemes and an exempt financial adviser for advising on, issuing and promulgating analyses and reports on securities. Sanford C. Bernstein (Singapore) Private Limited is registered in Singapore with Company Registration

No. 20213710W and located at 8 Marina Boulevard, #12-01, Marina Bay Financial Centre, Singapore 018981, +65-6326-7000.

To our readers in the People's Republic of China: The securities referred to in this document are not being offered or sold and may not be offered or sold, directly or indirectly, in the People's Republic of China (for such purposes, not including the Hong Kong and Macau Special Administrative Regions or Taiwan, the "PRC") in contravention of any applicable laws of the PRC.

This document does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC to any person to whom it is unlawful to make the offer or solicitation in the PRC.

We do not represent that this document may be lawfully distributed, or that any securities may be lawfully offered, in compliance with any applicable registration or other requirements in the PRC, or pursuant to an exemption available thereunder, or assume any responsibility for facilitating any such distribution or offering. In particular, no action has been taken by us which would permit a public offering of any securities or distribution of this document in the PRC. Accordingly, the securities are not being offered or sold within the PRC by means of this document or any other document. Neither this document nor any advertisement or other offering material may be distributed or published in the PRC, except under circumstances that will result in compliance with any applicable laws and regulations.

To our readers in Japan: This publication is being distributed in Japan by Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社), which is registered in Japan as a Financial Instruments Business Operator with the Kanto Local Finance Bureau (registration number: The Director-General of Kanto Local Finance Bureau (FIBO) No.3387) and regulated by the Financial Services Agency. It is also a member of Investment Management Association of Japan. This publication is solely for qualified institutional investors in Japan only, as defined in Article 2, paragraph (3), items (i) of the Financial Instruments and Exchange Act.

For the institutional client readers in Japan who have been granted access to the Bernstein website by Daiwa Securities Group Inc. ("Daiwa"), your access to this document should not be construed as meaning that Bernstein is providing you with investment advice for any purposes. Whilst Bernstein has prepared this document, your relationship is, and will remain with, Daiwa, and Bernstein has neither any contractual relationship with you nor any obligations towards you.

To our readers in Australia: Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is responsible for distributing research in Australia. It is regulated by the Securities and Exchange Commission under U.S. laws, by the Financial Conduct Authority under U.K. laws, which differs from Australian laws. Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is exempt from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the provision of the following financial services to wholesale clients:

- providing financial product advice;
- dealing in a financial product;
- making a market for a financial product; and
- providing a custodial or depository service.

To our readers in India: This publication is being distributed in India by Sanford C. Bernstein (India) Private Limited (SCB India) which is licensed and regulated by Securities and Exchange Board of India ("SEBI") as a research analyst entity under the SEBI (Research Analyst) Regulations, 2014, having registration no. INH000006378 and as a stock broker having registration no. INZ000213537. SCB India is currently engaged in the business of providing research and stock broking services. Please refer to www.bernsteinresearch.in for more information.

- SCB India is a Private limited company incorporated under the Companies Act, 2013, on April 12, 2017 bearing corporate identification number U65999MH2017FTC293762, and registered office at Level 3A, 4th Floor, First International Financial Centre, Plot Nos C-54 and C-55, G Block, Near CBI Office, Bandra Kurla Complex, Bandra (East), Mumbai 400098, Maharashtra, India (Phone No: +91-22-68421401).
- For details of Associates (i.e., affiliates/group companies) of SCB India, kindly email MUM-BERNSTEIN-InCompliance@bernsteinsg.com.
- SCB India does not have any disciplinary history as on the date of this report.
- Except as noted above, SCB India and/or its Associates (i.e., affiliates/group companies), the Research Analysts authoring this report, and their relatives

- do not have any financial interest in the subject company
- do not have actual/beneficial ownership of one percent or more in securities of the subject company;
- is not engaged in any investment banking activities for Indian companies, as such;
- have not managed or co-managed a public offering in the past twelve months for any Indian companies;
- have not received any compensation for investment banking services or merchant banking services from the subject company in the past 12 months;
- have not received compensation for brokerage services from the subject company in the past twelve months;
- have not received any compensation or other benefits from the subject company or third party related to the specific recommendations or views in this report; and
- do not currently, but may in the future, act as a market maker in the financial instruments of the companies covered in the report.
- do not have any conflict of interest in the subject company as of the date of this report.
- Except as noted above, the subject company has not been a client of SCB India during twelve months preceding the date of distribution of this research report. Neither SCB India nor its Associates (i.e., affiliates/group companies) have received compensation for products or services other than investment banking, merchant banking or brokerage services from the subject company in the past twelve months.
- The principal research analyst(s) who prepared this report, members of the analysts' team, and members of their households are not an officer, director, employee or advisory board member of the companies covered in the report.
- Our Compliance officer / Grievance officer is Ms. Rupal Talati, who can be reached at +91-22-68421451, or MUM-BERNSTEIN-InCompliance@bernsteinsg.com / Scbin-investorgrievance@bernsteinsg.com
- The Research investor charter and Terms & Conditions of SCB India are available on its website and may be accessed at [Sanford C. Bernstein \(India\) Private Limited](https://bernsteinresearch.in/) (<https://bernsteinresearch.in/>) for your reference.
- Disclaimer: Registration granted by SEBI, and certification from NISM, is in no way a guarantee of performance of the intermediary or provide any assurance of returns to investors. Investments in securities market are subject to market risks. Read all the related documents carefully before investing.

To our readers in Switzerland: This document is provided in Switzerland by or through Bernstein Autonomous LLP, and is provided only to qualified investors as defined in article 10 of the Swiss Collective Investment Scheme Act ("CISA") and related provisions of the Collective Investment Scheme Ordinance and in strict compliance with applicable Swiss law and regulations. The products mentioned in this document may not be suitable for all types of investors. This document is based on the Directives on the Independence of Financial Research issued by the Swiss Bankers Association (SBA) in January 2008.

To our readers in the Middle East: Bernstein Autonomous LLP, DIFC branch has its principal office at Gate Village 06, DIFC, Dubai, UAE. Bernstein Autonomous LLP, DIFC branch is regulated by the Dubai Financial Services Authority (DFSA) with the registration number CL10040 and is provisioned for Arranging Deals in Investments and Advising on Financial Products. All communications and services are directed at Professional Clients and Market Counterparties only (as defined in the DFSA rulebook). Persons other than Professional Clients and Market Counterparties, such as Retail Clients, are not the intended recipients of our communications or services.

LEGAL

All research publications are disseminated to our clients through posting on the firm's password protected websites, bernsteinresearch.com and autonomous.com. Certain, but not all, research publications are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience.

This publication has been published and distributed in accordance with the Firm's policy for management of conflicts of interest in investment research, a copy of which is available from Bernstein Institutional Services LLC, Director of Compliance, 245 Park

Avenue, New York, NY 10167. Additional disclosures and information regarding Bernstein's business are available on our website www.bernsteinresearch.com.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. This publication is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of, or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject any of the entities referenced herein or any of their subsidiaries or affiliates to any registration or licensing requirement within such jurisdiction. This publication is based upon public sources we believe to be reliable, but no representation is made by us that the publication is accurate or complete. We do not undertake to advise you of any change in the reported information or in the opinions herein. This publication was prepared and issued by entity referred to herein for distribution to eligible counterparties or professional clients. This publication is not an offer to buy or sell any security, and it does not constitute investment, legal or tax advice. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with their professional advisors in light of their specific circumstances. The value of investments may fluctuate, and investments that are denominated in foreign currencies may fluctuate in value as a result of exposure to exchange rate movements. Information about past performance of an investment is not necessarily a guide to, indicator of, or assurance of, future performance.

This report is directed to and intended only for our clients who are "eligible counterparties", "professional clients", "institutional investors" and/or "professional investors" as defined by the aforementioned regulators, and must not be redistributed to retail clients as defined by the aforementioned regulators. Retail clients who receive this report should note that the services of the entities noted herein are not available to them and should not rely on the material herein to make an investment decision. The result of such act will not hold the entities noted herein liable for any loss thus incurred as the entities noted herein are not registered/ authorised/ licensed to deal with retail clients and will not enter into any contractual agreement/arrangement with retail clients. This report is provided subject to the terms and conditions of any agreement that the clients may have entered into with the entities noted herein. All research reports are disseminated on a simultaneous basis to eligible clients through electronic publication to our client portal.

The information in this report was prepared by Bernstein solely for the internal business use of our clients. Clients may store, display, analyze, reformat and print the information in this report for this limited use only. Clients may not copy, alter, create derivative works, resell, reverse engineer, commercially exploit, share or distribute any part of the information contained herein for any purpose without Bernstein's express written consent. These restrictions include extracting data or using the content to develop indices or other products. Further, you may not use this report, or any portion of this report, to train or finetune any third-party machine learning or artificial intelligence system, or as a prompt or input into any such system. You also may not, without Bernstein's express written consent, do any of the foregoing in connection with your own internal machine learning or artificial intelligence system.

Bernstein may use artificial intelligence tools in the preparation of its materials. Any such materials are reviewed by Bernstein's research analysts prior to publication.

This report has been prepared for information purposes only and is based on current public information that we consider reliable, but the entities noted herein do not warrant or represent (express or implied) as to the sources of information or data contained herein are accurate, complete, not misleading or as to its fitness for the purpose intended even though the entities noted herein rely on reputable or trustworthy data providers, it should not be relied upon as such. Opinions expressed are the author(s)' current opinions as of the date appearing on the material only and we do not undertake to advise you of any change in the reported information or in the opinions herein.

Any references to Societe Generale herein are purely factual, based upon publicly available information, and included for comparative purposes only. They do not constitute an opinion or recommendation with respect to the securities of Societe Generale.

This publication was prepared and issued by the entity referred to herein for distribution to eligible counterparties or professional clients. The information in this report is intended for general circulation and does not constitute an offer to buy or sell any security, investment, legal or tax advice nor a personal recommendation, as defined by any of the aforementioned regulators. It does not take into account the particular investment objectives, financial situations, or needs of individual investors. The report has not been reviewed by any of the aforementioned regulators and does not represent any official recommendation from the aforementioned regulators. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with advice sought from a financial adviser regarding the suitability of the investment product, taking into account the specific investment objectives, financial situation or particular needs of any recipient of the recommendation, before the recipient makes a commitment to purchase the investment product.

The analysis contained herein is based on numerous assumptions. Different assumptions could result in materially different results. The information in this report does not constitute, or form part of, any offer to sell or issue, or any offer to purchase or

subscribe for shares, or to induce engagement in any other investment activity. The value of any securities or financial instruments mentioned in this report may fluctuate subject to market conditions. Information about past performance of an investment is not necessarily a guide to, indicative of, or assurance of future performance. Estimates of future performance mentioned by the research analyst in this report are based on assumptions that may not be realized due to unforeseen factors like market volatility/fluctuation. In relation to securities or financial instruments denominated in a foreign currency other than the clients' home currency, movements in exchange rates will have an effect on the value, either favorable or unfavorable. Before acting on any recommendations in this report, recipients should consider the appropriateness of investing in the subject securities or financial instruments mentioned in this report and, if necessary, seek independent professional advice.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject the entities noted herein to any regulation or licensing requirement within such jurisdiction.

Source: Bloomberg Index Services Limited. BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg"). Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Neither Bloomberg nor Bloomberg's licensors approves or endorses this material, or guarantees the accuracy or completeness of any information herein, or makes any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum extent allowed by law, neither shall have any liability or responsibility for injury or damages arising in connection therewith.

No part of this material may be reproduced, distributed or transmitted or otherwise made available without prior consent of the entities noted herein. Copyright Bernstein Institutional Services LLC Bernstein Autonomous LLP, BSG France S.A., Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited and Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社). All rights reserved. The trademarks and service marks contained herein are the property of their respective owners. Any unauthorized use or disclosure is strictly prohibited. The entities noted herein may pursue legal action if the unauthorized use results in any defamation and/or reputational risk to the entities noted herein and research published under the Bernstein and Autonomous brands.